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A mathematical model of dynamics of non-isothermal phase separation

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A mathematical model of thermally activated phase separation in binary systems is proposed. The construction is based on the Landau–Ginzburg theory of phase transitions and non-equilibrium thermodynamics. The proposed system of differential equations represents an extension of the Cahn–Hilliard model to the non-isothermal situation. The model is shown to be consistent with the first principles of thermodynamics. The stability of stationary solutions is considered, and for the one-dimensional case numerical results are presented.

1. Introduction

Dynamical processes of phase separation and related spatial pattern formation are characteristic for multi-component alloys, polymers, glasses, magnets and liquid mixtures (see [14,16]). Typically, binary systems under quenching are brought from one-phase equilibrium to non-equilibrium states which are located within the coexistence region of their phase diagrams. Then activated by composition fluctuations the systems evolve to a new equilibrium that comprises two coexisting phases. This contributes to a local separation of phases and creation of a spatially heterogeneous structure.

A phenomenological theory of phase separation has been proposed by Cahn and Hilliard [3–6]. Their model assumed that the process, caused by instantaneous quench below a critical temperature, is isothermal. Hence, only diffusive phenomena were taken into account.

Mathematical properties of the Cahn–Hilliard model were studied by Novick–Cohen and Segel [20,21], Elliott and Zheng [10,12,24]. Besides, the model was extensively studied by numerical simulation, see Elliott [10] and the references cited therein, Elliott and French [11]. Miyazaki et al. [18,19], and Swanger et al. [23].

In realistic physical systems quenches are usually carried out over a finite period of time. In such conditions phase separation can begin before the final quenching is reached. The consequences of this are to be accounted for, especially at early stages of the process and for systems with rapid diffusion time scale, see Cahn [4]. Besides, external thermal activation itself can be used to control the separation process and the resulting spatial structure. Huston et al. [15] studied phase separation during continuous cooling by considering the Cahn–Hilliard equation with temperature dependent coefficients.

Stochastic models of phase separation with

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thermally driven concentration fluctuations were developed by Cook [8] and Langer [17]. Stochastic aspects of periodic spinodal decomposition were studied by Onuki [22].

It is our purpose to propose a mathematical model of phase separation that describes coupled phenomena of mass diffusion and heat conduction in binary systems under thermal activation. The model extends the Cahn–Hilliard approach to the non-isothermal situation. The construction is based on the Landau–Ginzburg free energy functional and on non-equilibrium thermodynamics (see section 2). The proposed model has the form of a system of nonlinear parabolic differential equations for the concentration and the energy as conserved quantities. The model is conformable with the first principles of thermodynamics. Moreover, it has the relaxational property in the sense that there exists a Lyapunov functional non-increasing in time. The form of this functional is related to the energy conservation and entropy principle (see section 3). In the one-dimensional case we present a finite dimensional approximation of the model and a method of its numerical solution (see section 4). The corresponding results of numerical simulation are reported for the cases of both isothermal and non-isothermal phase separation (see section 5). Questions concerning existence of solutions to the model are considered separately in [2].

2. Phase separation in binary systems

2.1. Some phenomenology

We consider a binary system $\{A, B\}$ occupying a spatial domain $\Omega \subset \mathbb{R}^3$. Let θ denote the Kelvin temperature, c_A and c_B be the local concentrations of components A and B, scaled so that $c_A + c_B = 1$. Further we shall use $c = c_A$ as a system's variable. The corresponding specific free energy f alternates between qualitatively different forms as a function of c at various ranges of θ (see fig. 1a). At high temperatures θ, f is convex in c reflecting a continuous solubility range of the system. When passing the critical temperature θ_{crit}, a miscibility gap arises, and f assumes a double-well form. The phase diagram in fig. 1b corresponds to the latter form of f. The locus of concentration values c_{e_1}, c_{e_2}, defined by the supporting tangent to f, determines the coexistence curve in the (θ,c)-plane, referred to as the binodal. Such concentrations have identical chemical potentials and may coexist. The region above the binodal represents stable single phase states, its complement refers to states that are not thermodynamically stable. The inflection points c_{s_1}, c_{s_2} of f determine a curve referred to as the spinodal. This curve separates the metastable and unstable subregions with $f_{,cc} > 0$ and $f_{,cc} < 0$, respectively.

Initially in a homogeneous state, the system quenched below a critical temperature undergoes separation into two phases. For inner states of the spinodal, the path to phase separation is traditionally classified as spinodal decomposition, whereas in the metastable region as nucleation.

2.2. The Cahn–Hilliard model

First let us recall basic components of the Cahn-Hilliard model of isothermal phase separation. For isotropic heterogeneous binary systems with constant molar volume the total Helmholtz free energy is postulated in the form

$$F_\Omega(c) = \int_\Omega F(c)\,, \tag{2.1}$$

with the energy density

$$F(c) = f(c) + \tfrac{1}{2}\kappa|\nabla c|^2\,,$$

which comprises the volumetric free energy density $f(c)$ for the concentration c and the gradient term with a positive constant κ, representing an interfacial energy between the phases. Quali-

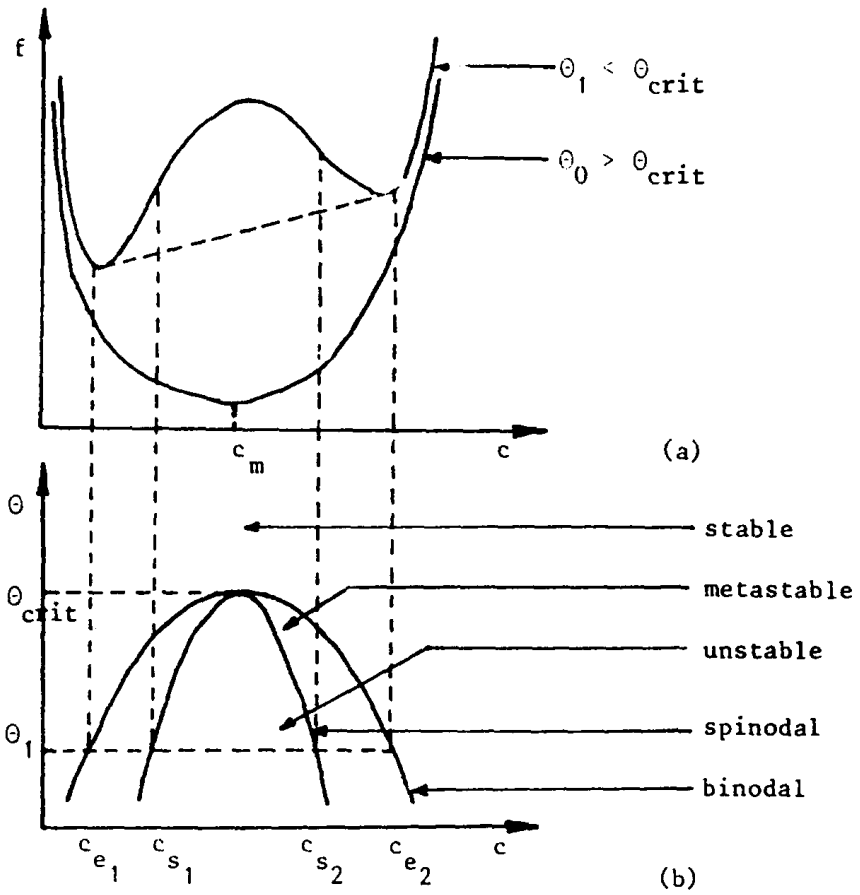


Fig. 1. (a) Free energy f as a function of c at temperatures $\theta_0 > \theta_{crit}$ and $\theta_1 < \theta_{crit}$. (b) Phase diagram of f as a function of c and θ.

tatively representative forms of f are shown in fig. 1. If the system is isolated at the boundary its isothermal equilibria are given by the stationary points of (2.1) subject to the additional constraint on the mean value of the concentration

$$\int_\Omega c(x)\,dx = c_m\,, \tag{2.2}$$

with a given constant c_m. This constraint means that the total mass of components A and B in the sample Ω is prescribed.

Therefore stationary solutions of (2.1) under constraint (2.2) must satisfy the corresponding Euler–Lagrange equation

$$f_{,c}(c) - \kappa\,\Delta c = \sigma\,, \quad \text{in } \Omega\,,$$

subject to the natural boundary condition

$$\nabla c \cdot \boldsymbol{n} = 0\,, \quad \text{on } \Gamma = \partial\Omega\,, \tag{2.3}$$

where $\boldsymbol{n}$ denotes the outward unit normal to Γ. Here σ is the Lagrange multiplier corresponding to constraint (2.2), given by

$$\sigma = \frac{\delta}{\delta c} F(c) = \int_\Omega f_{,c}(c)\,,$$

where $(\delta/\delta c)F$ denotes the first variation of the free energy functional F_Ω, which for a perturbation of c in the direction of a test function ζ is defined by

$$\left\langle \zeta, \frac{\delta}{\delta c} F(c) \right\rangle := \frac{d}{d\epsilon} F_\Omega(c + \epsilon\zeta)\Big|_{\epsilon=0}\,.$$

Hence, for local perturbation ζ we have

$$\left\langle \zeta, \frac{\delta}{\delta c} F(c) \right\rangle = \int_\Omega \zeta\,\frac{\delta}{\delta c} F(c)\,,$$

where

$$\frac{\delta}{\delta c} F(c) = f_{,c}(c) - \nabla \cdot (\kappa \nabla c), \quad \text{in } \Omega .$$

The above representation is also relevant for perturbations near the boundary Γ provided condition (2.3) is satisfied there.

We remark that the gradient term in (2.1) has a smoothing effect on interfaces between different phases. In consequence, jumps of c are not allowed, instead different phases are separated by walls, that is, small subregions with rapid changes of c. The thickness of these walls is related to the value of κ. It is well-known that in the limit $\kappa \to 0$ the surface area of the interface is minimized locally in Ω.

The dynamics of phase separation is described as follows. Let $c(t)$ be the concentration at time t, and μ be the difference between the chemical potentials of components A and B. It is defined as the first variation of the free energy functional

$$\mu = \mu_A - \mu_B := \frac{\delta}{\delta c} F(c), \tag{2.4}$$

that is,

$$\mu = f_{,c} - \kappa \Delta c . \tag{2.5}$$

Let $j = j_A$ and j_B denote the mass fluxes of A and B. It is assumed that $j_A + j_B = 0$, and generalized Fick's law

$$j = -M \nabla \mu \tag{2.6}$$

is postulated, where $M > 0$ is a parameter denoting the diffusive mobility. The mass balance law

$$\partial_t c + \nabla \cdot j = 0 \tag{2.7}$$

together with (2.5) and (2.6), provided $M =$ const. and $\kappa =$ const., yields the Cahn–Hilliard equation

$$\partial_t c - \nabla \cdot [M f_{,cc}(c) \nabla c] + M\kappa \Delta^2 c = 0, \quad \text{in } \Omega . \tag{2.8}$$

For this equation the natural boundary condition (2.3) is imposed together with the condition of mass isolation, that is,

$$\nabla c \cdot \boldsymbol{n} = 0, \quad j \cdot \boldsymbol{n} = 0, \quad \text{on } \Gamma . \tag{2.9}$$

The diffusion coefficient $D := M f_{,cc}(c)$ in (2.8) is both c- and θ-dependent. Below the critical temperature θ_{crit} it becomes negative at $c_{s_1} < c < c_{s_2}$ (see fig. 1), contributing then to an unstable development of the process.

We also note that the total free energy F_Ω acts as a Lyapunov functional for the Cahn-Hilliard equation, since

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\Omega(c(t)) \leq 0$$

for solutions of (2.8), (2.9) (see [12]).

2.3. *Model of non-isothermal phase separation*

The applicability range of the Cahn–Hilliard model is limited by the assumed isothermal character of the process, hence focusing only on concentration field. We shall extend the Cahn–Hilliard model by accounting for the dynamics of energy transfer. The model is formulated in terms of the Kelvin temperature θ and the concentration c as an order parameter.

Let $\Omega \subset \mathbb{R}^n$ ($n \leq 3$ in physical situations) be an open bounded domain with a smooth boundary Γ, and $]0, t_0[\subset \mathbb{R}_+ :=]0, \infty[$ be a finite time interval.

The total free energy of the binary system is assumed in the form of the Landau-Ginzburg potential with the energy density F given by (see [13])

$$F(c, \theta) = f(c, \theta) + \tfrac{1}{2}\kappa(c, \theta)|\nabla c|^2 \tag{2.10}$$

for functions c and θ. Here f: $\mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ is the volumetric energy density of a homogeneous system and κ: $\mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}_+$ a positive surface tension coefficient possibly dependent on the concentration and temperature. Further we shall

understand F as an operator mapping $C^1(\Omega) \times C^0(\Omega)$ into $C^0(\Omega)$. Let E denote the internal energy density of the system and S its entropy density. The Gibbs relation

$$F = E - \theta S$$

is postulated, with the entropy

$$S = -F_{,\theta} .$$

We generalize definition (2.4) of μ to the non-isothermal situation by introducing the reduced chemical potential difference

$$\frac{\mu}{\theta} := \frac{\delta}{\delta c}\left(\frac{F(c,\theta)}{\theta}\right).$$

This definition corresponds to the total Helmholtz free energy

$$\Phi_\Omega(c,\theta) := \int_\Omega \frac{F(c,\theta)}{\theta} ,$$

which means that for spatially non-uniform temperature fields θ the free energy density is scaled by $1/\theta$. For F as in (2.10) this gives

$$\frac{\mu}{\theta} = \frac{f_{,c}}{\theta} + \frac{1}{2}\frac{\kappa_{,c}}{\theta}|\nabla c|^2 - \nabla\cdot\left(\frac{\kappa\nabla c}{\theta}\right).$$

Mass balance is governed by eq. (2.7), and energy balance by

$$\partial_t E + \nabla\cdot \boldsymbol{q} = g$$

in $\Omega_{t_0} :=]0, t_0[\times \Omega$, where $\boldsymbol{q}$ is the energy flux (due to heat and mass transfer), and $g = g(t,x)$ represents the rate of distributed heat sources.

By employing the non-equilibrium thermodynamics formalism (see [9]), we postulate the constitutive relations for the mass and energy fluxes in the form

$$j = -l_{11}\nabla\left(\frac{\mu}{\theta}\right) + l_{12}\nabla\left(\frac{1}{\theta}\right),$$

$$\boldsymbol{q} = +l_{22}\nabla\left(\frac{1}{\theta}\right) - l_{21}\nabla\left(\frac{\mu}{\theta}\right),$$

with coefficients $l_{ij} = l_{ij}(c, \mu/\theta, 1/\theta)$ satisfying

$$l_{11} > 0 , \quad l_{22} > 0 , \quad l_{11}l_{22} - l_{12}l_{21} > 0 .$$

The model is completed by appropriate initial and boundary conditions for c and θ. At initial time $t = 0$,

$$c(0,x) = c_0(x) , \quad \theta(0,x) = \theta_0(x) , \quad \text{for } x \in \Omega , \tag{2.11}$$

with prescribed mean value

$$\fint_\Omega c_0(x)\,dx = c_m .$$

We assume that the system is thermodynamically closed, that is, it may exchange only heat with its environment. Then the condition of mass isolation

$$j\cdot n = 0 , \quad \text{on } \Gamma_{t_0} :=]0, t_0[\times \Gamma , \tag{2.12}$$

is satisfied. Besides, for the concentration condition (2.3) is imposed. For the temperature we postulate Newton's heat exchange through the boundary,

$$\boldsymbol{q}\cdot n + p\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = 0 , \quad \text{on } \Gamma_{t_0} , \tag{2.13}$$

where, in particular,

$$p\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = -p_0 \int_{\theta_{ext}}^{\theta} k\left(c, \frac{\mu}{\theta'}, \frac{1}{\theta'}\right) d\theta' - g_b . \tag{2.14}$$

The data above represent correspondingly, external temperature θ_{ext}, non-negative heat exchange coefficient $p_0(x)$, boundary flux $g_b(t,x)$, and the effective heat conductivity $k(c, \mu/\theta, 1/\theta)$ at the boundary. According to the constitutive

relations for the fluxes and by (2.12), k admits the representation

$$k\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = \frac{1}{\theta^2} l\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right),$$

$$\text{with } l = \frac{l_{11}l_{22} - l_{12}l_{21}}{l_{11}}.$$

We now can summarize the equations governing the dynamics of non-isothermal phase separation in the form of the following system in Ω_{t_0}:

$$-\frac{\mu}{\theta} + \frac{f_{,c}}{\theta} + \frac{1}{2}\frac{\kappa_{,c}}{\theta}|\nabla c|^2 - \nabla\cdot\left(\frac{\kappa\nabla c}{\theta}\right) = 0, \quad (2.15)$$

$$\partial_t c - \nabla\cdot\left[l_{11}\nabla\left(\frac{\mu}{\theta}\right) - l_{12}\nabla\left(\frac{1}{\theta}\right)\right] = 0, \quad (2.16)$$

$$-\partial_t E - \nabla\cdot\left[l_{22}\nabla\left(\frac{1}{\theta}\right) - l_{21}\nabla\left(\frac{\mu}{\theta}\right)\right] + g = 0, \quad (2.17)$$

where

$$E = e + \tfrac{1}{2}(\kappa - \theta\kappa_{,\theta})|\nabla c|^2, \quad e := f - \theta f_{,\theta},$$

together with initial conditions (2.11) at $t = 0$, and the boundary conditions on Γ_{t_0}

$$\nabla c\cdot\boldsymbol{n} = 0, \quad (2.18)$$

$$\left[-l_{11}\nabla\left(\frac{\mu}{\theta}\right) + l_{12}\nabla\left(\frac{1}{\theta}\right)\right]\cdot\boldsymbol{n} = 0, \quad (2.19)$$

$$\left[l_{22}\nabla\left(\frac{1}{\theta}\right) - l_{21}\nabla\left(\frac{\mu}{\theta}\right)\right]\cdot\boldsymbol{n} + p\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = 0. \quad (2.20)$$

Remark. There is a direct correspondence between the above system and the Cahn–Hilliard equation. Assume that $\kappa = \text{const.}$ and $l_{12} = 0$. Then if $l_{22} = \text{const.} \to \infty$, we arrive at

$$\Delta\left(\frac{1}{\theta}\right) = 0, \quad \text{in } \Omega,$$

$$\nabla\left(\frac{1}{\theta}\right)\cdot\boldsymbol{n} = 0, \quad \text{on } \Gamma.$$

This implies that $\theta = \theta(t)$ is a function of time only. Hence we get the Cahn–Hilliard equation with coefficients dependent on time (see [15]). If $\theta = \text{const.}$ our system reduces to eq. (2.8) with $M = l_{11}/\theta$.

Further we reformulate the above problem in terms of new variables

$$u := c - c_r, \quad v := \frac{\mu}{\theta}, \quad w := \frac{1}{\theta},$$

which represent the fluctuation of the concentration around some reference value c_r, the reduced chemical potential, and the inverse temperature, respectively. We also introduce the reduced free energy density

$$\Phi(u, w) := wF\left(c_r + u, \frac{1}{w}\right).$$

By definition of F,

$$\Phi(u, w) = \varphi(u, w) + \tfrac{1}{2}\kappa\left(c_r + u, \frac{1}{w}\right)w|\nabla u|^2,$$

$$\text{with } \varphi(u, w) := wf\left(c_r + u, \frac{1}{w}\right).$$

By the identity

$$\theta f_{,\theta} + w f_{,w} = 0,$$

the internal energy E and the entropy S, expressed in terms of Φ, are

$$E = \Phi_{,w} = e + \tfrac{1}{2}(\kappa w)_{,w}|\nabla u|^2, \quad e = \varphi_{,w},$$

$$S = w\Phi_{,w} - \Phi = s + \tfrac{1}{2}w^2\kappa_{,w}|\nabla u|^2,$$

$$s = w\varphi_{,w} - \varphi. \quad (2.21)$$

Then we come to the following setting of the model problem in Ω_{t_0}:

$$-v + \varphi_{,u} + \tfrac{1}{2}\kappa_{,u}w|\nabla u|^2 - \nabla\cdot(\kappa w\nabla u) = 0, \quad (2.22)$$

$$\partial_t u - \nabla\cdot(l_{11}\nabla v - l_{12}\nabla w) = 0\,, \tag{2.23}$$

$$-\partial_t E - \nabla\cdot(l_{22}\nabla w - l_{21}\nabla v) + g = 0\,, \tag{2.24}$$

together with the initial conditions

$$u(0,x) = u_0(x)\,,\quad w(0,x) = w_0(x)\,,\quad \text{for } x\in\Omega\,, \tag{2.25}$$

and the boundary conditions on Γ_{t_0}

$$\nabla u\cdot n = 0\,, \tag{2.26}$$

$$(l_{11}\nabla v - l_{12}\nabla w)\cdot n = 0\,, \tag{2.27}$$

$$(l_{22}\nabla w - l_{21}\nabla v)\cdot n + p(u,v,w) = 0\,, \tag{2.28}$$

where $u_0 = c_0 - c_r$ and $w_0 = 1/\theta_0$, so that

$$\fint_\Omega u_0(x)\,dx = c_m - c_r =: u_m\,.$$

3. Main structural properties of the model

3.1. Thermodynamical compatibility

The model (2.22)–(2.28) (in terms of the primary variables, (2.15)–(2.20)) conforms with two basic laws of thermodynamics. Indeed, by integrating (2.23) and (2.24) over Ω and using boundary conditions (2.27), (2.28) we get the identities

$$\frac{d}{dt}\int_\Omega u = 0\,,$$

$$\frac{d}{dt}\int_\Omega E(u,w) = \int_\Omega g + \int_\Gamma p(u,v,w)\,d\mathcal{H}^{n-1}\,,$$

which correspondingly express conservation of mass and energy.

Also the second principle of thermodynamics expressed by the Clausius–Duhem inequality for the entropy production holds in the integrated form over Ω, that is,

$$\frac{d}{dt}\int_\Omega S + \int_\Omega [\nabla\cdot(wq - vj) - wg] \geq 0\,. \tag{3.1}$$

To obtain this inequality we multiply eq. (2.23) by v and (2.24) by w, then integrate over Ω using (2.27) and (2.28). Since the matrix $(l_{ij})_{i,j=1,2}$ in the constitutive relations for the fluxes is positive definite, we get

$$0 \geq \int_\Omega (v\,\partial_t u - w\,\partial_t E) + \int_\Omega wg + \int_\Gamma wp(u,v,w)\,d\mathcal{H}^{n-1}\,. \tag{3.2}$$

After substituting v from (2.22) and using condition (2.26), the first integral in the above inequality is

$$\int_\Omega [(\varphi_{,u} + \tfrac{1}{2}\kappa_{,u} w|\nabla u|^2)\,\partial_t u + \tfrac{1}{2}\kappa w\,\partial_t|\nabla u|^2 + \Phi_{,w}\,\partial_t w - \partial_t(w\Phi_{,w})] = \int_\Omega \partial_t(\Phi - w\Phi_{,w}) = -\int_\Omega \partial_t S\,.$$

In view of conditions (2.27) and (2.28) the boundary integral in (3.2) is equal to

$$-\int_\Gamma (wq - vj)\cdot n\,d\mathcal{H}^{n-1}\,.$$

Hence Clausius–Duhem inequality (3.1) follows.

3.2. Basic assumptions

The existence of weak solutions to the problem (2.22)–(2.28) has been proved in [2]. We outline the main assumptions under which the existence result is valid.

3.2.1. Interfacial parameter κ

$\kappa = \text{const.} > 0$.

3.2.2. Free energy φ

$\varphi : \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$

is of class C^2, and admits a splitting

$\varphi = \varphi^0 + \varphi^1$,

with C^2-functions φ^0 and φ^1. In the sequel (u, w) denotes any element of $\mathbb{R} \times \mathbb{R}_+$. The following convexity properties are postulated as the main structural conditions on φ:

$$-\varphi_{,ww}(u, w) \geq \omega_0(w) > 0 , \tag{3.3}$$

with a continuous function $\omega_0 : \mathbb{R}_+ \mapsto \mathbb{R}_+$,

$\varphi^0_{,uu}(u, w) \geq 0$,

$\pm \varphi^0_{,u}(u, w) \to +\infty$, for $\pm u \to +\infty$.

Besides, growth conditions on φ are imposed:

$|\varphi^0_{,u}(u, w)| \leq C(1 + w)(1 + |u|^{\nu - 1})$

for some exponent $\nu > 2$, and

$|\varphi^1_{,u}(u, w)|, |\varphi^1_{,uu}(u, w)| \leq C(1 + w)$,

$|\varphi^1_{,uw}(u, w)| \leq C$.

These conditions are compatible with the assumption that $\varphi(u, w)$ is concave with respect to w, which implies that $\varphi(u, w)$ in its positive part is sublinear in w.

Moreover we remark that assumption (3.3) implies that the constant-volume specific heat coefficient $C_V := e_{,\theta}$, which by Gibbs' relation equals

$C_V = -\theta f_{,\theta\theta} = -w^2 \varphi_{,ww}$,

is positive.

3.2.3. Internal energy e and entropy s

The subsequent structural assumptions concern internal energy e and entropy s. If $\kappa =$ const., then by (2.21) the entropy includes only the volumetric part, that is, $S = s$. Then the corresponding assumptions read as follows.

For some positive constants α_0, c, C,

$$\alpha_0 e(u, w) - s(u, w) \geq c[\psi_0(u) + \psi_1(w)] - C , \tag{3.4}$$

where

$\psi_0 \geq 0$, $\quad \dfrac{\psi_0(u)}{|u|} \to \infty$, as $|u| \to \infty$,

and ψ_0 is monotone for large values of $|u|$,

$\psi_1 \geq 0$, $\quad \psi_1(w) \to \infty$, as $w \to 0$.

In addition to (3.4) appropriate estimates on the internal energy and the entropy are postulated. In particular they imply that the main terms of $e(u, w)$ are positive, and comprise $\psi_0(u)$ and $\psi_1(w)$, compare the example in section 3.3. On the other hand, they imply that the entropy does not contain the main terms of the energy.

3.2.4. Elliptic coefficients

$l_{ij} : \mathbb{R} \times \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$, $\quad i, j = 1, 2$,

are continuous globally bounded functions, and the matrix

$$L = \begin{pmatrix} l_{11} & -l_{12} \\ -l_{21} & l_{22} \end{pmatrix}$$

is strictly positive definite, that is, for some $c > 0$

$\xi \cdot L(u, v, w)\xi \geq c|\xi|^2$, $\quad$ for $\xi \in \mathbb{R}^2$.

3.2.5. The source g and boundary term p

$g \in L^\infty(\Omega_{t_0})$;

$p: \mathbb{R} \times \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ is a continuous function satisfying

$$cp_0(w) - C \le p(u, v, w) \le C[p_0(w) + 1],$$

where c and C are positive constants, and p_0: $\mathbb{R}_+ \to \mathbb{R}_+$ is a continuous function such that

$$p_0(w) \to \infty, \quad \text{as } w \to \infty,$$
$$0 \le p_0(w) \le w,$$
$$0 \le w p_0'(w) \le C[p_0(w) + 1]. \tag{3.5}$$

The condition (3.5), necessary for mathematical considerations in [2], means that for very low temperatures a large portion of heat is supplied to the system. It is to be pointed out, however, that since the actual applicability range of the model is restricted to some finite range of temperatures around the critical point θ_{crit}, (3.5) is only of mathematical importance.

3.3. Example of free energy

We give an example of the free energy f that matches the above structural assumptions. Assume f in the generalized Landau form

$$f(c, \theta) = \delta_V f_*(\theta) + f_1(\theta) f_2(c) + f_3(c), \tag{3.6}$$

where f_*, f_i are given functions and δ_V is a positive constant. The first term in (3.6) represents the main concave term of the free energy. It refers to pure heat conduction and, in particular, can be postulated in one of the forms

$$f_*(\theta) = \delta_1 \theta + \delta_2 - \theta \log \theta, \tag{3.7}$$

$$f_*(\theta) = -\theta^2, \tag{3.8}$$

$$f_*(\theta) = -\theta \log(1 + \theta), \tag{3.9}$$

where δ_1, $\delta_2 \ge 0$ are constants.

Then by (2.21), the corresponding parts of the reduced free energy φ_*, the internal energy e_* and the entropy s_* associated with f_* are

$$\varphi_*(w) = \delta_1 + \delta_2 w + \log w, \quad e_*(w) = \delta_2 + \frac{1}{w},$$
$$s_*(w) = -\delta_1 + 1 - \log w,$$

for f_* in (3.7),

$$\varphi_*(w) = -\frac{1}{w}, \quad e_*(w) = \frac{1}{w^2}, \quad s_*(w) = \frac{2}{w},$$

for f_* in (3.8), and

$$\varphi_*(w) = -\log\left(1 + \frac{1}{w}\right), \quad e_*(w) = \frac{1}{w(1 + w)},$$
$$s_*(w) = \frac{1}{1 + w} + \log\left(1 + \frac{1}{w}\right),$$

for f_* in (3.9).

The second term in (3.6) is in charge of changes in the qualitative behavior of f below a critical temperature θ_{crit} from the convex into non-convex double-well form (see fig. 1). The third term in (3.6) determines the behavior of f for large values of $|c|$. The expressions for f_i, phenomenologically justified for various solid solutions, for instance, metallic alloys or polymers, are usually fourth or sixth order polynomials in c (see [18,19,21]):

$$f_1(\theta) = \alpha_1(\theta - \theta_{crit}), \quad f_2(c) = c^2,$$
$$f_3(c) = -\alpha_3 c^3 + \alpha_4 c^4 + \alpha_6 c^6, \tag{3.10}$$

with constants θ_{crit}, $\alpha_1 > 0$, α_3, α_4, $\alpha_6 \ge 0$ and $\alpha_4 + \alpha_6 > 0$. Then $f(c, \theta)$ is convex in c for $\theta > \theta_{crit}$, and

$$\varphi(u, w) = \delta_V \varphi_*(w) + \alpha_1(1 - w\theta_{crit})(c_r + u)^2 + w f_3(c_r + u),$$
$$e(u, w) = \delta_V e_*(w) - \alpha_1 \theta_{crit}(c_r + u)^2 + f_3(c_r + u),$$
$$s(u, w) = \delta_V s_*(w) - \alpha_1(c_r + u)^2.$$

The expressions for the specific heat coefficient C_V associated with (3.7), (3.8) and (3.9) are, respectively,

$$C_V = \delta_V\,, \quad C_V = \delta_V\theta\,, \quad C_V = \delta_V\Big(1 - \frac{1}{(1+\theta)^2}\Big)\,.$$

We note also that for f_* given by (3.8) and (3.9) the third law of thermodynamics expressed as

$$s \to 0\,, \quad \text{as } \theta \to 0$$

is satisfied. All postulated structural assumptions on the free energy, internal energy and entropy are satisfied. The concavity condition (3.3) holds directly by the definition of f_*. To verify the other conditions on φ, we decompose

$$\hat f(c) := f_3(c) - \alpha_1\theta_{\text{crit}}c^2 = \hat f^0(c) + \hat f^1(c)\,,$$

with C^2-functions $\hat f^0$ and $\hat f^1$, so that $\hat f^0$ is convex and $\hat f^0(c) = \hat f(c)$ for large $|c|$. Then $\hat f^1$ has compact support, and the required growth conditions on φ follow by defining

$$f^0(c,\theta) := \delta_V f_*(\theta) + \alpha_1\theta c^2 + \hat f^0(c)\,,$$

$$f^1(c,\theta) := \hat f^1(c)\,,$$

and, correspondingly,

$$\varphi^i(u,w) := wf^i\Big(c_r + u, \frac{1}{w}\Big)\,, \quad i = 0,1\,.$$

The condition (3.4) on the internal energy and the entropy is satisfied with

$$\psi_0(u) := |u|^\nu\,,$$

where $\nu = 6$ if $\alpha_6 > 0$, and $\nu = 4$ if $\alpha_6 = 0$,

$$\psi_1(w) := e_*(w)\,.$$

3.4. Nonlinear stability

Here we discuss the stability of stationary solutions to the system (2.22)–(2.28). A corresponding Lyapunov functional, non-increasing on solution paths, is constructed, and a nonlinear stability analysis is performed in the one-dimensional case. A linear stability analysis for the system (2.22)–(2.28) is presented in section 3.5.

We consider stationary solutions $(\bar u, \bar v, \bar w)$ of system (2.22)–(2.28) in the special case with

$$\kappa = \text{const.} > 0\,,$$

$$l_{11} = l_{22} = 1\,, \quad l_{12} = l_{21} = 0 \quad \text{(no cross-effects)}\,,$$

$$p = p(w)\,, \quad \text{monotone increasing in } w.$$

Then it follows from (2.23) and (2.27) that

$$\bar v = \sigma = \text{const.}$$

Eqs. (2.22) and (2.24) with boundary conditions (2.26) and (2.28) then read

$$\begin{aligned} &\varphi_{,u}(\bar u, \bar w) - \kappa\,\nabla\cdot(\bar w\,\nabla\bar u) = \sigma\,, \quad \text{in } \Omega\,,\\ &\nabla\bar u\cdot\boldsymbol{n} = 0\,, \quad \text{on } \Gamma\,, \end{aligned} \tag{3.11}$$

$$\begin{aligned} &-\Delta\bar w + g = 0\,, \quad \text{in } \Omega\,,\\ &\nabla\bar w\cdot\boldsymbol{n} + p(\bar w) = 0\,, \quad \text{on } \Gamma\,. \end{aligned} \tag{3.12}$$

Also,

$$\fint_\Omega \bar u(x)\,\mathrm{d}x = u_m\,.$$

Let $(u_0 - \bar u, w_0 - \bar w)$ be a perturbation of the stationary state $(\bar u, \bar w)$ with $\fint_\Omega u_0(x)\,\mathrm{d}x = u_m$, and let (u, v, w) be a solution of the system (2.22)–(2.24) with boundary conditions (2.26)–(2.28) and initial data (u_0, w_0). Then

$$\begin{aligned} &v - \bar v = \varphi_{,u} - \kappa\,\nabla\cdot(w\,\nabla u) - \sigma\,, \quad \text{in } \Omega_{t_0}\,,\\ &\nabla(u - \bar u)\cdot\boldsymbol{n} = 0\,, \quad \text{on } \Gamma_{t_0}\,, \end{aligned} \tag{3.13}$$

$$\begin{aligned} &\partial_t(u - \bar u) - \Delta(v - \bar v) = 0\,, \quad \text{in } \Omega_{t_0}\,,\\ &\nabla(v - \bar v)\cdot\boldsymbol{n} = 0\,, \quad \text{on } \Gamma_{t_0}\,, \end{aligned} \tag{3.14}$$

$$-\partial_t(E-\bar{E})-\Delta(w-\bar{w})=0\,,\quad \text{in } \Omega_{t_0}\,,$$
$$\nabla(w-\bar{w})\cdot n+p(w)-p(\bar{w})=0\,,\quad \text{on } \Gamma_{t_0}\,. \tag{3.15}$$

Now we introduce the corresponding Lyapunov functional $Q_\Omega(u, w)$ which has the property

$$\frac{\mathrm{d}}{\mathrm{d}t}\,Q_\Omega(u(t), w(t))\leq 0\,.$$

To obtain this functional we multiply eq. (3.14) by $v-\bar{v}$ and eq. (3.15) by $w-\bar{w}$. Using (3.13) and (3.11) we get

$$\begin{aligned}(v-\bar{v})\,\partial_t(u-\bar{u})&-(w-\bar{w})\,\partial_t(E-\bar{E})\\ &=[(\varphi_{,u}-\bar{\varphi}_{,u})\,\partial_t(u-\bar{u})\\ &\quad-(w-\bar{w})\,\partial_t(\varphi_{,w}-\bar{\varphi}_{,w})]\\ &\quad+\kappa\{[-\nabla\cdot(w\,\nabla u)+\nabla\cdot(\bar{w}\,\nabla\bar{u})]\,\partial_t(u-\bar{u})\\ &\quad-\tfrac{1}{2}(w-\bar{w})\,\partial_t(|\nabla u|^2-|\nabla\bar{u}|^2)\}\,,\end{aligned} \tag{3.16}$$

where the notation $\bar{\varphi}=\varphi(\bar{u}, \bar{w})$ is used. After integrating over Ω, the κ-terms in this equality are

$$\kappa\,\partial_t\int_\Omega[\tfrac{1}{2}\bar{w}|\nabla u|^2-\bar{w}\,\nabla\bar{u}\cdot\nabla(u-\bar{u})]\,, \tag{3.17}$$

and the potential terms can be expressed as

$$\partial_t\int_\Omega[\bar{w}\varphi_{,w}-(u-\bar{u})\bar{\varphi}_{,u}+\varphi-w\varphi_{,w}]\,. \tag{3.18}$$

From (3.11) we have

$$\begin{aligned}-\partial_t&\int_\Omega(u-\bar{u})\bar{\varphi}_{,u}\\ &=-\partial_t\int_\Omega(u-\bar{u})\sigma+\kappa\,\partial_t\int_\Omega\nabla(u-\bar{u})\cdot(\bar{w}\,\nabla\bar{u})\,,\end{aligned}$$

where the first term on the right-hand side vanishes since u and $\bar{u}$ have equal mean values over Ω. Therefore (3.18) is

$$\begin{aligned}\partial_t&\int_\Omega[\bar{w}e(u, w)-S(u, w)]\\ &+\kappa\,\partial_t\int_\Omega\bar{w}\,\nabla\bar{u}\cdot\nabla(u-\bar{u})\,.\end{aligned}$$

Combining the latter expression with (3.17) we arrive at the identity

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\,Q_\Omega(u, w)&+\int_\Omega[|\nabla(v-\bar{v})|^2+|\nabla(w-\bar{w})|^2]\\ &+\int_\Gamma(w-\bar{w})[p(w)-p(\bar{w})]\,\mathrm{d}\mathcal{H}^{n-1}=0\,,\end{aligned} \tag{3.19}$$

where

$$Q_\Omega(u, w)=\int_\Omega Q(u, w)\,,$$
$$\text{with}\quad Q(u, w):=\bar{w}E(u, w)-S(u, w)\,.$$

Hence, Q_Ω is closely related to the energy conservation and the entropy principle (see section 3.1).

According to (3.11),

$$\frac{\delta}{\delta u}\,Q(\bar{u}, \bar{w})=\sigma\,,\quad \text{moreover}\ \frac{\delta}{\delta w}\,Q(\bar{u}, \bar{w})=0\,.$$

This means that $(\bar{u}, \bar{w})$ is a stationary point for the functional Q_Ω under the constraint $\fint_\Omega(u-\bar{u})=0$ (see section 2.1). We also note that Q_Ω can be represented as

$$Q_\Omega(u, w)=Q^0_\Omega(u)+Q^1_\Omega(u, w)\,,$$

where

$$Q^0_\Omega(u):=Q_\Omega(u, \bar{w})=\int_\Omega\varphi(u, \bar{w})+\tfrac{1}{2}\kappa\int_\Omega\bar{w}|\nabla u|^2$$

and

$$\begin{aligned}Q^1_\Omega(u, w):=\int_\Omega&[\varphi(u, w)-\varphi(u, \bar{w})\\ &-(w-\bar{w})\varphi_{,w}(u, w)]\,.\end{aligned}$$

The functional Q^0_Ω corresponds to the mass diffusion equation with a given, in general non-constant, temperature distribution $1/\bar{w}$ (if $p=0$ and $g=0$ then $\bar{w}=\text{const.}>0$, by (3.12)). Thus Q^0_Ω is the Lyapunov functional associated with the uncoupled Cahn–Hilliard equation. We have then

$$\frac{\delta}{\delta u} Q^0(\bar{u}, \bar{w}) = \frac{\delta}{\delta u} Q(\bar{u}, \bar{w}) = \sigma .$$

The functional Q^1_Ω refers to the coupled mass and heat equations. Since $w \mapsto \varphi(u, w)$ is concave by assumption (3.3), Q^1_Ω is non-negative.

We shall now study the connection of the Lyapunov functional Q_Ω to the stability of the stationary solution $(\bar{u}, \bar{w})$. For this purpose we define the distance of a pair (u, w) from $(\bar{u}, \bar{w})$ by

$$d(u, w) := d_0(u) + Q^1_\Omega(u, w) ,$$

where

$$\begin{aligned} d_0(u) := \int_\Omega & [\varphi^0(u, \bar{w}) - \varphi^0(\bar{u}, \bar{w}) \\ & - (u-\bar{u})\varphi^0_{,u}(\bar{u}, \bar{w})] \\ & + \tfrac{1}{2}\kappa \int_\Omega \bar{w}|\nabla(u-\bar{u})|^2 \geq 0 . \end{aligned}$$

Then

$$\begin{aligned} & Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w}) \\ & \quad = Q^1_\Omega(u, w) + \int_\Omega [\varphi(u, \bar{w}) - \varphi(\bar{u}, \bar{w})] \\ & \qquad + \tfrac{1}{2}\kappa \int_\Omega \bar{w}(|\nabla u|^2 - |\nabla \bar{u}|^2) . \end{aligned}$$

Using eq. (3.11) this can be expressed as

$$\begin{aligned} & d(u, w) + \int_\Omega (u-\bar{u}) \\ & \quad \times \int_0^1 [\varphi^1_{,u}((1-\alpha)\bar{u} + \alpha u, \bar{w}) - \varphi^1_{,u}(\bar{u}, \bar{w})] \, d\alpha . \end{aligned}$$

Therefore, by the growth condition on $\varphi^1_{,uu}$, we get the estimate

$$\begin{aligned} & |Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w}) - d(u, w)| \\ & \quad \leq C \int_\Omega |u-\bar{u}|^2(1+\bar{w}) . \end{aligned} \tag{3.20}$$

We say that the stationary solution $(\bar{u}, \bar{w})$ is stable if and only if for all sufficiently small perturbations $(u_0 - \bar{u}, w_0 - \bar{w})$ of $(\bar{u}, \bar{w})$, that is, if $d(u_0, w_0)$ is small enough, and satisfying $\int_\Omega (u_0 - \bar{u}) = 0$, we have

$$d(u(t), w(t)) \to 0 , \quad \text{as } t \to \infty ,$$

where $(u(t), w(t))$ is the solution of the system (2.22)–(2.24) with boundary conditions (2.26)–(2.28), and initial data (u_0, w_0).

Since by (3.19) the functional $Q_\Omega(u(t), w(t))$ is non-increasing in time it is natural to expect that the stability of $(\bar{u}, \bar{w})$ is related to the positivity of the quantities

$$\lambda_{\alpha,\rho} := \inf_{\alpha\rho \leq d(u,w) \leq \rho} [Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w})] ,$$

where $\rho > 0$ and $0 < \alpha < 1$.

We shall show that for a given α, and provided $\bar{w}$ is strictly positive and bounded, the following two statements are valid:

$$\begin{aligned} & \lambda_{\alpha,\rho} < 0 , \quad \text{for a sequence } \rho \to 0 , \\ & \quad \text{implies that } (\bar{u}, \bar{w}) \text{ is unstable,} \end{aligned} \tag{3.21}$$

$$\begin{aligned} & \lambda_{\alpha,\rho} > 0 , \quad \text{for a sequence } \rho \to 0 , \\ & \quad \text{implies that } (\bar{u}, \bar{w}) \text{ is stable.} \end{aligned} \tag{3.22}$$

To prove (3.21), let us fix $\rho > 0$ and choose (u_0, w_0) with $d(u_0, w_0) \leq \rho$ so that $Q_\Omega(u_0, w_0) < Q_\Omega(\bar{u}, \bar{w})$. Integrating (3.19) in time we obtain that for $t \geq 0$

$$Q_\Omega(u(t), w(t)) \leq Q_\Omega(u_0, w_0) < Q_\Omega(\bar{u}, \bar{w}) . \tag{3.23}$$

If $d(u(t), w(t)) \to 0$ for a sequence $t \to \infty$, it follows from the definition of d that $\nabla u(t) \to \nabla \bar{u}$ in $L^2(\Omega)$. Therefore (3.20) implies that $Q_\Omega(u(t), w(t)) \to Q_\Omega(\bar{u}, \bar{w})$, which is a contradiction to (3.23).

We shall now prove (3.22). To this purpose choose $\rho_0 > 0$ sufficiently small to assure that in the neighborhood $\{u; d(u, \bar{w}) \le \rho_0\}$ the function $\bar{u}$ is the only stationary solution of the Cahn–Hilliard equation with given temperature distribution $1/\bar{w}$. It is to be noted that in the one-dimensional case this is always possible since then the number of stationary solutions is finite (see Zheng [24]). Now take (u_0, w_0) so that $d(u_0, w_0) \le \delta_0 < \rho_0$. Then by (3.20),

$$\begin{aligned} & Q_\Omega(u_0, w_0) - Q_\Omega(\bar{u}, \bar{w}) \\ & \quad \le d(u_0, w_0) + C \int_\Omega |u_0 - \bar{u}|^2 \\ & \quad \le C d(u_0, w_0) \\ & \quad \le C\delta_0 < \lambda_{\alpha, \rho_0}, \end{aligned} \tag{3.24}$$

provided δ_0, depending on ρ_0, is chosen small enough. Using properties of solutions (u, v, w) to the system (2.22)–(2.24) established in [2], we infer from the identity (3.19) that $Q_\Omega(u(t), w(t))$ coincides almost everywhere with a continuous function of time. Besides, by the mass balance equation (2.23) it follows that $u(t)$ is a continuous function of time with values in $L^2(\Omega)$. Therefore, according to (3.20), also $d(u(t), w(t))$ coincides almost everywhere with a continuous function of time. Then it follows from (3.24) and the definition of λ_{α, ρ_0} that $d(u(t), w(t)) < \alpha\rho_0$ for all t. In addition, we infer from the identity (3.19) that

$$\begin{aligned} & \int_0^\infty \Big(\int_\Omega [|\nabla v(t) - \nabla \bar{v}|^2 + |\nabla w(t) - \nabla \bar{w}|^2] \\ & \quad + \int_\Gamma [w(t) - \bar{w}][p(w(t)) - p(\bar{w})] \, d\mathcal{H}^{n-1} \Big) dt \\ & \quad < \infty . \end{aligned}$$

With this estimate, in the one-dimensional case, by using growth conditions on $\varphi_{,u}$, we conclude that for a subsequence $t \to \infty$ the following holds: $w(t) \to \bar{w}$, $v(t) \to \sigma_\infty$ strongly in $H^{1,2}(\Omega)$ and $u(t) \to u_\infty$ weakly in $H^{1,2}(\Omega)$ with $\sigma_\infty \in \mathbb{R}$ and $u_\infty \in H^{1,2}(\Omega)$. Then we pass to the limit with $t \to \infty$ in the differential equations (2.22) and (2.23). This implies that $(u_\infty, \bar{w})$ is a stationary solution. Since $d(u_\infty, \bar{w}) < \alpha\rho_0$ due to the lower semicontinuity of d, and since $\bar{u}$ has been assumed to be an isolated stationary solution of the Cahn–Hilliard equation, we conclude that $u_\infty = \bar{u}$. This means that u_∞ is the stationary solution of the Cahn–Hilliard equation with temperature distribution $1/\bar{w}$. Moreover, we can prove that $\nabla u(t) \to \nabla \bar{u}$ strongly in $L^2(\Omega)$, therefore $d(u(t), w(t)) \to 0$ for a subsequence $t \to \infty$. Further asymptotic analysis using estimates on the modulus of continuity of u in time shows that $d(u(t), w(t)) \to 0$ for the whole sequence $t \to \infty$. This proves the stability statement (3.22) in the one-dimensional case.

It is of interest to discuss the relationship between the stability criteria for the system (2.22)–(2.24) and for the Cahn–Hilliard equation alone, that is, (2.22) and (2.23) with a given $\bar{w}$. The corresponding criterion is described by the quantity

$$\lambda^0_{\alpha, \rho} := \inf_{\alpha\rho \le d(u, \bar{w}) \le \rho} [Q_\Omega(u, \bar{w}) - Q_\Omega(\bar{u}, \bar{w})] .$$

Obviously, the stability considerations presented above apply also for the Cahn–Hilliard equation (formally, we set everywhere $w = \bar{w}$ and neglect all terms coming from the energy equation (2.24)). In particular, the stability criteria (3.21) and (3.22) remain valid for λ replaced by λ^0. Since

$$\begin{aligned} Q_\Omega(u, w) &= Q_\Omega(u, \bar{w}) + Q^1_\Omega(u, w) , \\ d(u, w) &= d(u, \bar{w}) + Q^1_\Omega(u, w) , \end{aligned}$$

we see that

$$\lambda_{\alpha, \rho} \le \lambda^0_{\alpha, \rho} .$$

This inequality means that if the solution $(\bar{u}, \bar{w})$ of the system is stable then also the solution $\bar{u}$ to the Cahn–Hilliard equation with a given $\bar{w}$ is stable. Conversely, if the solution $\bar{u}$ of the Cahn–Hilliard equation is unstable then also the corresponding solution $(\bar{u}, \bar{w})$ of the system is unstable.

As a matter of fact we are going to show the equivalence between the stability criteria (3.21), (3.22) for the system and the corresponding criteria for the Cahn–Hilliard equation alone. To this purpose we shall estimate $\lambda_{\alpha,\rho}$ from below. Let us choose a sufficiently small $\delta_0 > 0$, and consider two situations for (u, w) in the definition of $\lambda_{\alpha,\rho}$. If $d(u, \bar{w}) \geq \delta_0 \alpha\rho$ then, since $d(u, \bar{w}) \leq \rho$,

$$Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w}) \geq Q_\Omega(u, \bar{w}) - Q_\Omega(\bar{u}, \bar{w}) \geq \lambda^0_{\delta_0\alpha,\rho} .$$

Otherwise, $d(u, \bar{w}) \leq \delta_0 \alpha\rho$. Now by (3.20), provided $\bar{w}$ is bounded from below and from above by positive constants, we have

$$Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w}) \geq d(u, w) - C \int_\Omega |u - \bar{u}|^2 \geq d(u, w) - C_0 d(u, \bar{w}) ,$$

with some constant $C_0 > 0$. Therefore,

$$Q_\Omega(u, w) - Q_\Omega(\bar{u}, \bar{w}) \geq \alpha\rho - C_0 d(u, \bar{w}) \geq \alpha\rho(1 - C_0\delta_0) \geq \tfrac{1}{2}\alpha\rho ,$$

provided δ_0 is chosen sufficiently small. Simultaneously, by similar arguments,

$$\lambda^0_{\delta_0\alpha,\rho} \leq \inf_{d(u,\bar{w}) = \delta_0\alpha\rho} [Q_\Omega(u, \bar{w}) - Q_\Omega(\bar{u}, \bar{w})] \leq \inf_{d(u,\bar{w}) = \delta_0\alpha\rho} (1 + C_0) d(u, \bar{w}) = (1 + C_0)\delta_0\alpha\rho \leq \tfrac{1}{2}\alpha\rho ,$$

provided δ_0 is chosen small enough.

Combining the above considerations we conclude that for sufficiently small $\delta_0 > 0$, independent of α and ρ,

$$\lambda_{\alpha,\rho} \geq \lambda^0_{\delta_0\alpha,\rho} .$$

This estimate implies the following conclusions: If the stability criterion (3.22) with $\alpha < \delta_0$ is satisfied for the Cahn–Hilliard equation with given $\bar{w}$ then this criterion with α/δ_0 holds also for the system. Besides, if the instability criterion (3.21) with $\alpha > 0$ is satisfied for the system then it is also satisfied with $\delta_0\alpha$ for the Cahn–Hilliard equation.

In the one-dimensional case on a finite space interval, stability questions for the Cahn–Hilliard equation have been recently studied in [20,21], also [7]. Results on the asymptotic long time behavior of solutions to the Cahn–Hilliard equation have been obtained in [24]. We recall that in the one-dimensional case, and if $p = 0$, $g = 0$, the stationary solution $\bar{u}$ satisfies

$$\frac{\kappa\bar{w}}{2}\left(\frac{d\bar{u}}{dx}\right)^2 = \varphi(\bar{u}, \bar{w}) - (\sigma\bar{u} + \tau)$$

for some constant τ. It is well-known that there are three classes of stationary solutions: periodic, non-periodic and van der Waal's solution (see [21]). The number of solutions has been estimated in [24]. In fig. 2 periodic solutions correspond to the situation in which the line $l(u) := (1/\bar{w})(\sigma u + \tau)$ intersects $f(u, \bar{w})$ with two positive angles. The non-periodic solutions (sometimes referred to as nucleation solutions) are related to the case of l being tangential to f on one side. Van der Waal's solution refers to the situation in which l is the supporting tangent to f. This solution corresponds to global separation of the system into two phases. As pointed out in section 2.2, these stationary solutions are critical points of Q^0_Ω subject to the constraint

$$\fint_\Omega u(x)\, dx = u_m ,$$

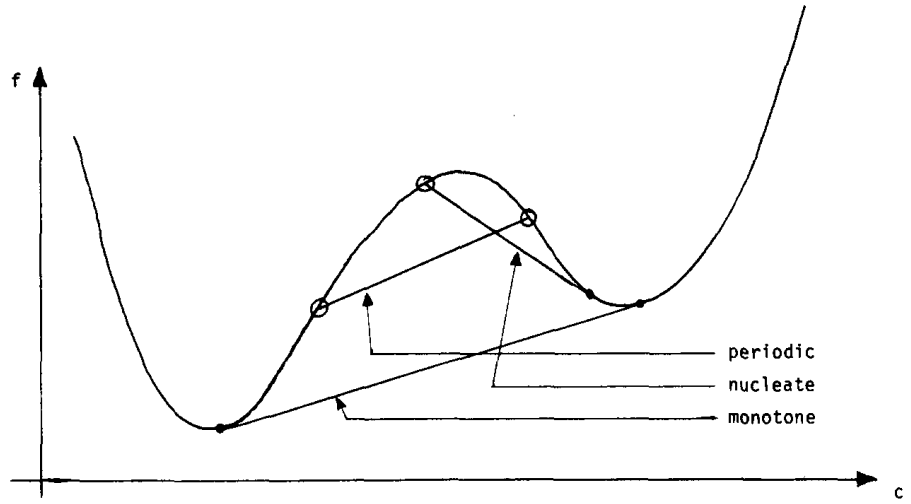


Fig. 2. Illustration of one-dimensional stationary solutions.

with σ being the corresponding Lagrange multiplier. It has been proved (see [7]) that there exists a unique global minimizer for the functional Q_Ω^0 with this constraint and that it is strictly monotone, thus corresponding to the single-interface solution. Besides, all non-monotone solutions of the stationary problem, that is, solutions with more than one interface, are unstable in the sense that they cannot be even local minimizers of Q_Ω^0 with the above constraint.

3.5. Linear stability

For a linear stability analysis to the Cahn–Hilliard equation we refer in particular to [3] and [17]. Here we are concerned with a linear stability analysis for stationary solutions of the system (2.22)–(2.28). By introducing a quadratic expansion of the functional Q_Ω around a stationary solution $(\bar u, \bar w)$ we obtain the following approximate counterpart $\tilde Q_\Omega$ of the functional Q_Ω:

$$\tilde Q_\Omega(u, w) := \tfrac{1}{2}\int_\Omega [\bar\varphi_{,uu}(u-\bar u)^2 + \kappa\bar w|\nabla(u-\bar u)|^2 - \bar\varphi_{,ww}(w-\bar w)^2]\,. \tag{3.25}$$

The corresponding linear version of eqs. (3.13)–(3.15) for perturbations $\tilde u := u - \bar u$, $\tilde v := v - \bar v$, and $\tilde w := w - \bar w$ is

$$\tilde v = \bar\varphi_{,uu}\tilde u + \bar\varphi_{,uw}\tilde w - \kappa\,\nabla\cdot(\bar w\,\nabla\tilde u) - \kappa\,\nabla\cdot(\tilde w\,\nabla\bar u)\,, \quad \text{in } \Omega_{t_0}\,,$$
$$\nabla\tilde u\cdot\boldsymbol{n} = 0\,, \quad \text{on } \Gamma_{t_0}\,, \tag{3.26}$$

$$\partial_t\tilde u - \Delta\tilde v = 0\,, \quad \text{in } \Omega_{t_0}\,,$$
$$\nabla\tilde v\cdot\boldsymbol{n} = 0\,, \quad \text{on } \Gamma_{t_0}\,, \tag{3.27}$$

$$-\partial_t\tilde E - \Delta\tilde w = 0\,, \quad \text{in } \Omega_{t_0}\,,$$
$$\nabla\tilde w\cdot\boldsymbol{n} + \bar p'\tilde w = 0\,, \quad \text{on } \Gamma_{t_0}\,, \tag{3.28}$$

where

$$\tilde E = \bar\varphi_{,ww}\tilde w + \bar\varphi_{,wu}\tilde u + \kappa\,\nabla\bar u\cdot\nabla\tilde u\,, \tag{3.29}$$

and the notation $\bar\varphi = \varphi(\bar u, \bar w)$, $\bar p' = p'(\bar w)$, etc. is used. The initial conditions are

$$\tilde u(0, x) = u_0(x) - \bar u(x)\,,$$
$$\tilde w(0, x) = w_0(x) - \bar w(x)\,,$$

and the mass constraint is $\fint_\Omega \tilde u(0, x)\,dx = 0$.

Clearly, $\tilde Q_\Omega$ given by (3.25) acts as the Lyapunov functional for the above linearized system. We note that if $\bar w = \text{const.}$ and $\tilde w = 0$,

and if the stationary solution $\bar{u} = \text{const.}$, then (3.27) reduces to the equation

$$\partial_t \tilde{u} - D_0 \Delta \tilde{u} + M\kappa \Delta^2 \tilde{u} = 0 , \tag{3.30}$$

with constant coefficients $D_0 = M f_{,uu}(\bar{u}, \bar{w})$ and $M = \bar{w}$. It is usually referred to as linearized Cahn–Hilliard equation (see (2.8)).

Here we briefly recall the basic predictions of the linear stability analysis of eq. (3.30) (see [3,4]). We shall refer to this in section 5 where numerical results are discussed. In the one-dimensional case (3.30) admits a solution

$$\tilde{u}(t, x) = A(\beta, t) \cos \beta x , \tag{3.31}$$

where

$$A(\beta, t) = A(\beta, 0) \exp[R(\beta) t] ,$$

$$R(\beta) = -\beta^2 (D_0 + M\kappa\beta^2) .$$

The representation (3.31) for perturbations $\tilde{u}$ provides an explicit characterization of the development in time of initial cosine-type fluctuations of the concentration. If $D_0 > 0$ then $R(\beta)$ is negative, and thus the system remains stable with respect to fluctuations of any wavelength with any initial amplitude $A(\beta, 0)$. Consequently, this results in a decay of all spatial heterogeneities. If $D_0 < 0$, then there exist critical parameters, a critical wavenumber β_c and a corresponding critical wavelength λ_c, given by

$$\beta_c = \sqrt{-\frac{D_0}{M\kappa}} , \quad \lambda_c = \frac{2\pi}{\beta_c} , \tag{3.32}$$

such that $R(\beta_c) = 0$. For wavelengths $\lambda > \lambda_c$ the characteristic number $R(\beta)$ is positive and attains its maximum at

$$\lambda_{\text{opt}} = \sqrt{2}\, \lambda_c . \tag{3.33}$$

Therefore, at negative D_0, the system becomes unstable with respect to infinitesimal cosine fluctuations of wavelengths larger than λ_c.

To observe explicitly the impact of the coupling of the Cahn–Hilliard equation with the energy equation on the stability criterion, let us consider the eigenvalue problem associated with the linearized system (3.26)–(3.28). Assuming that $\bar{u} = \text{const.}$ and $\bar{w} = \text{const.} > 0$ we have

$$\partial_t \tilde{u} - a \Delta \tilde{u} + b \Delta^2 \tilde{u} - c \Delta \tilde{w} = 0 ,$$

$$d\, \partial_t \tilde{w} - \Delta \tilde{w} - c\, \partial_t \tilde{u} = 0 ,$$

where

$$a := \bar{\varphi}_{,uu} , \quad b := \kappa \bar{w} > 0 ,$$

$$c := \bar{\varphi}_{,uw} , \quad d := -\bar{\varphi}_{,ww} > 0 .$$

A one-dimensional solution $(\tilde{u}, \tilde{w})$ to this system is given by $\tilde{u}$ as in (3.31), and

$$\tilde{w}(t, x) = B(\beta, t) \cos \beta x ,$$

satisfying

$$\frac{\mathrm{d}}{\mathrm{d}t} \begin{pmatrix} A(\beta, t) \\ B(\beta, t) \end{pmatrix} + P(\beta) \begin{pmatrix} A(\beta, t) \\ B(\beta, t) \end{pmatrix} = 0 ,$$

where

$$P(\beta) := \beta^2 \begin{pmatrix} a + b\beta^2 & c \\ (c/d)(a + b\beta^2) & (1 + c^2)/d \end{pmatrix} .$$

The eigenvalues λ_+ and λ_- of the matrix $P(\beta)$ are given by (we assume that $\lambda_- \le \lambda_+$)

$$\mu_\pm := \frac{2\lambda_\pm}{\beta^2} = W \pm \sqrt{Z^2 + \frac{4c^2}{d} (a + b\beta^2)} \tag{3.34'}$$

$$= W \pm \sqrt{W^2 - \frac{4}{d} (a + b\beta^2)} , \tag{3.34''}$$

where

$$W := (a + b\beta^2) + \left(\frac{1 + c^2}{d} \right) ,$$

$$Z := (a + b\beta^2) - \left(\frac{1+c^2}{d}\right),$$

$$W + Z = 2(a + b\beta^2) =: \mu_{ch},$$

$$\mu_{ch} = \frac{2\lambda_{ch}}{\beta^2}, \quad \lambda_{ch} := \beta^2(a + b\beta^2).$$

λ_{ch} is the corresponding eigenvalue for the Cahn–Hilliard equation (3.30) with $a = D_0$ and $b = \kappa M$. We note that the stability condition $R(\beta) < 0$ for eq. (3.30) reads equivalently as $\lambda_{ch} > 0$. Using the two representations in (3.34) for the eigenvalues $\mu_\pm$ we can make the following observations:

Let $a + b\beta^2 > 0$. Then by (3.34′),

$$\mu_- < W - |Z| \le W + Z = \mu_{ch},$$

and by (3.34″),

$$\mu_- > W - |W| = 0.$$

In turn, assuming that $a + b\beta^2 < 0$, it follows from (3.34′) that

$$\mu_- > W - |Z| = W + Z = \mu_{ch},$$

and by (3.34″)

$$\mu_- < W - |W| \le 0.$$

By the above expressions we conclude the following:

(1) The instability condition for the system, $\mu_- < 0$, is satisfied if and only if it is satisfied for the Cahn–Hilliard equation, $\mu_{ch} < 0$.

(2) If the Cahn–Hilliard equation is stable, that is, $\mu_{ch} > 0$, then the system is less stable, $0 < \mu_- < \mu_{ch}$. On the contrary, if the Cahn–Hilliard equation is unstable, that is, $\mu_{ch} < 0$, then the system is less unstable, $\mu_{ch} < \mu_- < 0$.

4. Method of numerical solution

We now construct a finite-dimensional scheme approximating the model (2.22)–(2.28), and describe a numerical method for solving the resulting discrete problem in the case of one space dimension.

Let the time interval $[0, t_0]$ be uniformly divided into N subintervals $[t_i, t_{i+1}]$, $i = 0, 1, \ldots, N-1$, with $t_i = i\tau$, where τ is the time step. We shall use the notation

$$u^i(x) := u(t_i, x), \quad u(x) := u(t_{i+1}, x), \quad \text{etc.},$$

with the upper index $i+1$ omitted everywhere for simplicity. The system (2.22)–(2.24) is discretized in time by a fully implicit scheme.

We assume that the free energy f has the form (3.6), (3.7), and the boundary term p is given by (2.14). Then the corresponding time-discrete problem admits the following weak formulation:

$$-(v, \zeta)_{L^2(\Omega)} + (\kappa w \nabla u, \nabla\zeta)_{L^2(\Omega)} + (\Phi_{,u}, \zeta)_{L^2(\Omega)} = 0, \tag{4.1}$$

$$\frac{1}{\tau}(u, \xi)_{L^2(\Omega)} + (l_{11}\nabla v - l_{12}\nabla w, \nabla\xi)_{L^2(\Omega)} = \frac{1}{\tau}(u^i, \xi)_{L^2(\Omega)}, \tag{4.2}$$

$$-\frac{\delta_V}{\tau}\left(\frac{1}{w}, \eta\right)_{L^2(\Omega)} + (l_{22}\nabla w - l_{21}\nabla v, \nabla\eta)_{L^2(\Omega)} + (p, \eta)_{L^2(\Gamma)} = (R_\Omega(u, w), \eta)_{L^2(\Omega)}, \tag{4.3}$$

for all $\zeta, \xi, \eta \in H^{1,2}(\Omega)$, $i = 1, \ldots, N$,

where

$$\Phi_{,u}(u, w) = w(f_1 f_{2,c} + f_{3,c}) + \tfrac{1}{2} w\kappa_{,u}|\nabla u|^2,$$

$$R_\Omega(u, w) = -g + \frac{1}{\tau}[-E^i + \tfrac{1}{2}(\kappa w)_{,w}|\nabla u|^2 + \delta_V \cdot \delta_2 + (wf_1)_{,w} f_2 + f_3].$$

The nonlinear terms $\Phi_{,u}$ and p on the left-hand sides of (4.1) and (4.3) undergo a special treatment. Since we are interested in solutions with

concentration values in the non-convex part of the potential, we replace $\Phi_{,u}$ for large values of u by a linear function of u. Assuming that $f(c, \theta)$ is convex for large values of $|c|$, we can choose values $c_1(\theta) < c_2(\theta)$ so that

$$f_{,c}(c, \theta) < 0\,, \quad \text{for } c \le c_1(\theta)\,,$$

$$f_{,c}(c, \theta) > 0\,, \quad \text{for } c > c_2(\theta)\,.$$

Outside the interval $[c_1(\theta), c_2(\theta)]$ we extrapolate $f_{,c}$ by the linear function

$$q(c, \theta) = a(\theta)c + b(\theta)\,,$$

where

$$a(\theta) = \frac{f_{,c}(c_2(\theta)) - f_{,c}(c_1(\theta))}{c_2(\theta) - c_1(\theta)} > 0\,,$$

$$b(\theta) = \frac{f_{,c}(c_1(\theta))\, c_2(\theta) - f_{,c}(c_2(\theta))\, c_1(\theta)}{c_2(\theta) - c_1(\theta)}\,.$$

Accordingly, we replace $\Phi_{,u}(u, w)$ by

$$wa\left(\frac{1}{w}\right) u - R(u, w)\,,$$

where

$$R(u, w) = \begin{cases} -wq(c_r, 1/w) - \frac{1}{2} w\kappa_{,u}(c_r + u, 1/w)|\nabla u|^2 \\ \quad \text{for } u \notin [c_1(1/w) - c_r, c_2(1/w) - c_r]\,, \\ -\Phi_{,u}(u, w) + wa(1/w)u\,, \\ \quad \text{otherwise}\,. \end{cases}$$

Using (2.14) we split the boundary term p into

$$p\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = p_0 l\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) \frac{1}{\theta} - R_\Gamma\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right),$$

with

$$R_\Gamma\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) = p_0 l\left(c, \frac{\mu}{\theta}, \frac{1}{\theta}\right) \frac{1}{\theta_{\text{ext}}}$$

$$+ p_0 \int_{\theta_{\text{ext}}}^{\theta} \frac{l(c, \mu/\theta', 1/\theta') - l(c, \mu/\theta, 1/\theta)}{\theta'^2}\, \mathrm{d}\theta' + g_b\,.$$

Then (4.1)–(4.3) become

$$\frac{1}{\tau}\left(\left[wa\left(\frac{1}{w}\right)u - v\right], \zeta\right)_{L^2(\Omega)} + \frac{1}{\tau}(\kappa w \nabla u, \nabla\zeta)_{L^2(\Omega)} = \frac{1}{\tau}(R(u, w), \zeta)_{L^2(\Omega)}\,, \tag{4.4}$$

$$\frac{1}{\tau}(u, \xi)_{L^2(\Omega)} + (l_{11} \nabla v, \nabla\xi)_{L^2(\Omega)} = \frac{1}{\tau}(u^i, \xi)_{L^2(\Omega)} + (l_{12} \nabla w, \nabla\xi)_{L^2(\Omega)}\,, \tag{4.5}$$

$$-\frac{\delta_V}{\tau}\left(\frac{1}{w}, \eta\right)_{L^2(\Omega)} + (l_{22} \nabla w, \nabla\eta)_{L^2(\Omega)} + (p_0 l w, \eta)_{L^2(\Gamma)} = (R_\Omega(u, w), \eta)_{L^2(\Omega)} + (l_{21} \nabla v, \nabla\eta)_{L^2(\Omega)} + (R_\Gamma(u, v, w), \eta)_{L^2(\Gamma)}\,. \tag{4.6}$$

For the numerical treatment of these equations we used finite dimensional subspaces V_m as Galerkin approximation of $H^{1,2}(\Omega)$. We implemented this in the one-dimensional case with V_m consisting of piecewise linear elements over a uniformly discretized space interval $\Omega =]0, L[$. The discrete problem (4.4)–(4.6) was solved by a two-stage iterative procedure:

(i) for fixed w, system (4.4), (4.5) was solved with respect to u and v;

(ii) for fixed u, v, eq. (4.6) was solved with respect to w.

Let us denote by $(\underline{u}, \underline{v}, \underline{w})$ the vector of coefficients of (u, v, w) with respect to a given basis of V_m. To perform (i) we note that the left-hand sides of (4.4), (4.5) are linear in u and v, except that l_{11} and κ might depend explicitly on one of these variables. Therefore we write eqs. (4.4) and (4.5) as

$$D(\underline{u}^{(\nu)}, \underline{w})\underline{u}^{(\nu+1)} - B\underline{v}^{(\nu+1)} = \underline{F}(\underline{u}^{(\nu)}, \underline{v}^{(\nu)}, \underline{w})\,,$$

$$B\underline{u}^{(\nu+1)} + A(\underline{u}^{(\nu)}, \underline{v}^{(\nu)}, \underline{w})\underline{v}^{(\nu+1)} = \underline{G}(\underline{u}^{(\nu)}, \underline{w})\,.$$

Here $\underline{w}$ is a given temperature distribution for step (i), and $(\underline{u}^{(\nu)}, \underline{v}^{(\nu)})$ denotes the νth iterate to solve (4.4), (4.5). In our implementation we used a linear solver in the form of Gauss elimination with 2×2-matrices as entries.

Because of the above truncation of $\Phi_{,u}$ we had to check that in course of computations the bounds $c_1(1/w) - c_r$ and $c_2(1/w) - c_r$ were not exceeded by the actual u-values. If necessary we extended the interval $[c_1(\theta), c_2(\theta)]$ for the u-values.

To perform step (ii) we used an iteration procedure for (4.6). The $(\nu+1)$th iterate $w^{(\nu+1)}$ was determined from the following equation:

$$
\begin{aligned}
&-\frac{\delta_V}{\tau}\left(\frac{1}{w^{(\nu+1)}}, \eta\right)_{L^2(\Omega)} \\
&+ (l_{22}(u, v, w^{(\nu)})\, \nabla w^{(\nu+1)}, \nabla \eta)_{L^2(\Omega)} \\
&+ (p_0 l(u, v, w^{(\nu)}) w^{(\nu+1)}, \eta)_{L^2(\Gamma)} \\
&= (R_\Omega(u, w^{(\nu)}), \eta)_{L^2(\Omega)} \\
&\quad + (l_{21}(u, v, w^{(\nu)})\, \nabla v, \nabla \eta)_{L^2(\Omega)} \\
&\quad + (R_\Gamma(u, v, w^{(\nu)}), \eta)_{L^2(\Gamma)} \,. \qquad (4.7)
\end{aligned}
$$

This equation for $w^{(\nu+1)}$ was solved by means of the corresponding minimum problem for the functional

$$
\begin{aligned}
J^{(\nu)}(w) := &-\frac{\delta_V}{\tau}\int_\Omega \log w \\
&+ \tfrac{1}{2}\int_\Omega l_{22}(u, v, w^{(\nu)})\, |\nabla w|^2 \\
&+ \tfrac{1}{2}\int_\Gamma p_0 l(u, v, w^{(\nu)})\, w^2 \, d\mathcal{H}^{n-1} \,,
\end{aligned}
$$

which is convex in w. Adding to $J^{(\nu)}(w)$ a linear term corresponding to the right-hand side of (4.7) we solved the minimization problem in the space V_m by a nonlinear SOR-procedure. The main advantage of this method is that the first term of $J^{(\nu)}$ gives the positivity of $w^{(\nu+1)}$.

Therefore, given values u^i, w^i at time moment t_i, a sequence of approximations to a solution (u, v, w) of the discrete problem (4.4)–(4.6) at time moment t_{i+1} was generated iteratively, according to the procedure described above.

5. Numerical simulation

We report numerical experiments performed with the use of the proposed model of non-isothermal phase separation in the one-dimensional case and for constant surface tension coefficient κ. Of our special interest were dynamical aspects of structure formation in the separation process under thermal activation. First we present results referring to a situation in which the system is instantaneously quenched below a critical temperature and then kept under isothermal conditions. In such a case our model reduces to the Cahn–Hilliard equation. That equation was extensively studied by numerical methods, see [10] and references cited therein, and [11,18,19,23]. In the isothermal case we focus on the impact of the mean concentration c_m on the phase separation development. The results of non-isothermal simulation apply to systems subject to boundary cooling. In this case we are interested in the influence of cooling conditions on the phase separation, on one hand, and on the thermal effects associated with spatial pattern formation, on the other.

5.1. Free energy

The volumetric free energy f is assumed symmetric with respect to $c = \frac{1}{2}$,

$$
f(c, \theta) = \delta_V f_*(\theta) + f_1(\theta)\,(c - \tfrac{1}{2})^2 + \alpha_6 (c - \tfrac{1}{2})^6 \,, \qquad (5.1)
$$

with f_* given by (3.7), and

$$
f_1(\theta) = \alpha_1(\theta - \theta_{crit}) \,, \quad \alpha_1, \alpha_6 > 0 \,.
$$

The above form is phenomenologically justified for some solid solutions (cf. [18,19]). In this

case, for $\theta < \theta_{crit}$ the binodal concentrations $c_{e_i}(\theta)$, $i = 1,2$, satisfy

$$[c_{e_i}(\theta) - \tfrac{1}{2}]^4 = -\frac{f_1(\theta)}{3\alpha_6}\,, \tag{5.2}$$

while the spinodal concentrations $c_{s_i}(\theta)$ are related to $c_{e_i}(\theta)$ by

$$5[c_{s_i}(\theta) - \tfrac{1}{2}]^4 = [c_{e_i}(\theta) - \tfrac{1}{2}]^4\,. \tag{5.3}$$

In the numerical experiments we have assumed different mean values $c_m \le \frac{1}{2}$. To highlight the role of c_m, we express the diffusion coefficient $D := Mf_{,cc}$ of the Cahn–Hilliard equation (2.8) in terms of fluctuations of the concentration around the mean value, that is, $u = c - c_m$ (thus, $c_r = c_m$). For the free energy (5.1), this gives

$$D(u, \theta) = D_0(\theta) + (c_m - \tfrac{1}{2})D_1(u)\,u + D_2u^4\,, \tag{5.4}$$

where

$$D_0(\theta) = 2Mf_1(\theta) + D_2(c_m - \tfrac{1}{2})^4$$
$$= -D_2\{[c_{s_i}(\theta) - \tfrac{1}{2}]^4 - (c_m - \tfrac{1}{2})^4\}\,,$$

$$D_1(u) = 4D_2\{[u + \tfrac{3}{4}(c_m - \tfrac{1}{2})]^2 + \tfrac{7}{16}(c_m - \tfrac{1}{2})^2\}\,,$$

$$D_2 = 30\alpha_6 M\,.$$

Note that D_0 corresponds to the linearized Cahn–Hilliard equation (3.30). It is negative for c_m inside the spinodal, $c_{s_1}(\theta) < c_m \le \frac{1}{2}$, vanishes at the spinodal, $c_m = c_{s_1}(\theta)$, and is positive outside the spinodal, $c_m < c_{s_1}(\theta)$. The second term in (5.4) vanishes for $c_m = \frac{1}{2}$, and its linear coefficient $(c_m - \frac{1}{2})D_1(0)$ increases in c_m.

5.2. Isothermal case

First we present results referring to the isothermal situation. Values of parameters in the numerical experiments have been chosen so as to be representative for some solid solutions, with the diffusion data taken from Miyazaki et al. [18,19]. In scaled units, that is, length 10^{-9} m, time 10^3 s, energy 10^{-12} J, and temperature K, these data are

$$\theta = 1000\,,$$
$$L = 78.15\,, \quad T = 2000\,,$$
$$\alpha_1 = 2.1 \times 10^{-4}\,, \quad \alpha_6 = 0.61\,,$$
$$\kappa = 0.815 \times 10^{-2}\,, \quad l_{11} = 10^3\,, \quad l_{12} = 0\,.$$

Temperature is assumed constant, $\theta = 1000$, below the critical point $\theta_{crit} = 1500$. According to (5.2) and (5.3), the corresponding spinodal values are then $c_{s_1} \approx 0.17$, $c_{s_2} \approx 0.83$ and the binodal values $c_{e_1} \approx 0.01$, $c_{e_2} \approx 0.99$.

The function u_0, representing a small initial perturbation around a constant value c_m is assumed in the form

$$u_0(x) = 0.005 \sum_{i=2}^{5} \gamma_i(-1)^{i+1} \cos\left(i\,\frac{2\pi x}{L}\right), \tag{5.5}$$

with $\gamma_i = 1$ for $i < 5$ and $\gamma_5 = 4$.

The results of experiments displayed in figs. 3–6 show the influence of the mean concentration c_m on the dynamics of phase separation and the development of the resulting pattern. Fig. 3 refers to the binary system with equi-phase composition, $c_m = \frac{1}{2}$. Fig. 3a displays the early time development of the initial concentration fluctuation, and fig. 3b its continuation over a later time interval. After some time, the initial fluctuation produces a uniform cosine pattern with a period corresponding to the fastest wavelength. The amplitude of the wave grows in time whereas its period remains approximately unchanged. Such process is referred to as granulation, with concentration extrema interpreted as phase grains. The pattern in fig. 3a remains in qualitative agreement with the solution (3.31). Actually, as follows from (5.4), early stages of the process (corresponding to small u) with $c_m = \frac{1}{2}$ are primarily controlled by the D_0-term, thus can be well described by the linearized Cahn–

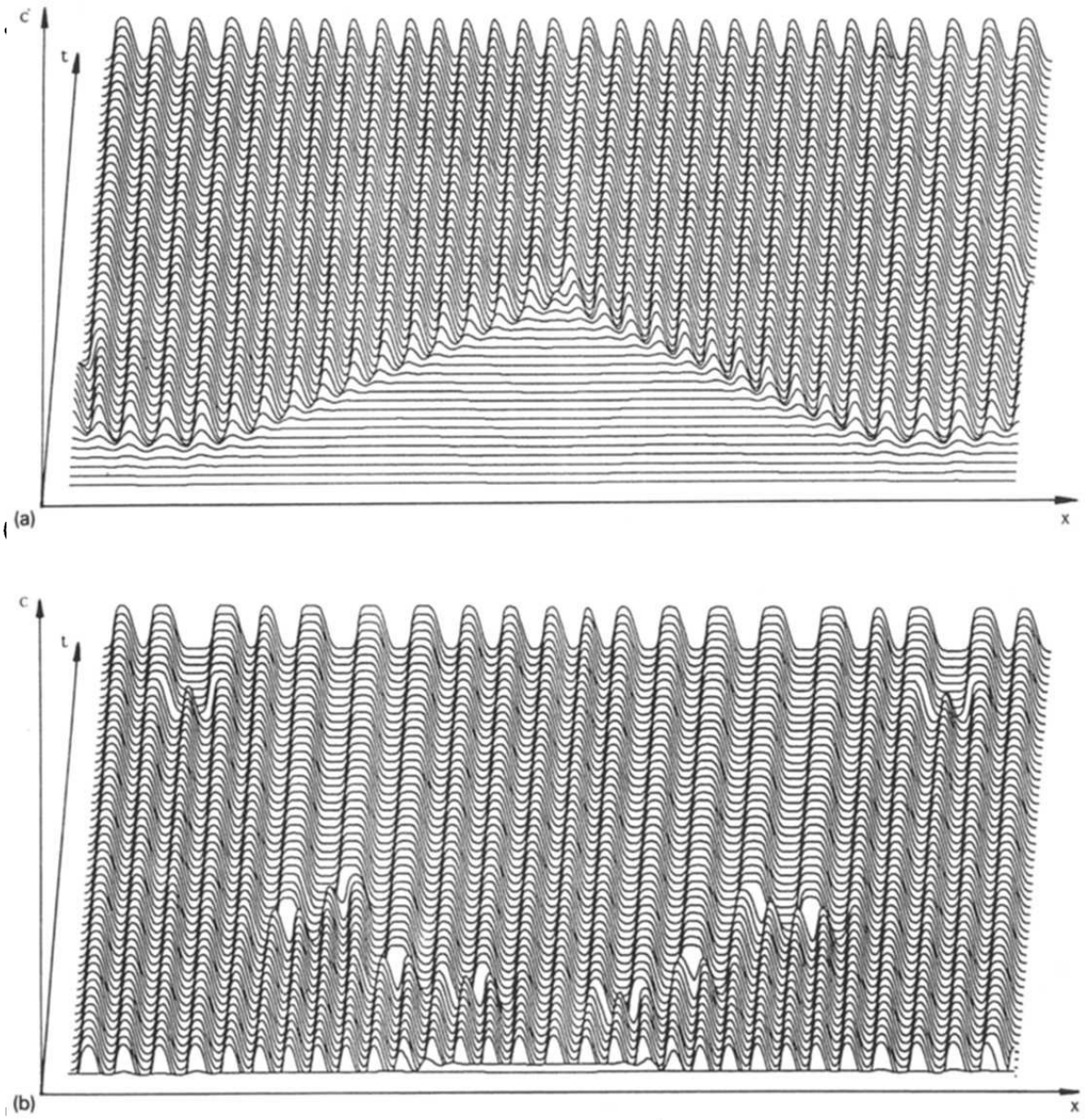


Fig. 3.

Hilliard equation. We observe in fig. 3b that during process development some of the peaks decay. This phenomenon, referred to as grain (or zone) aggregation, is characteristic of the later stages of phase separation (see [18,19]). Due to competitive growth of the grains, some of them, developed at early stages of the process, are later annihilated. As a consequence large grains develop on the cost of their finer neighbours. This reflects a tendency of the process to forming a periodic phase pattern with wavelength possibly large. When the amplitude of the peaks achieves the binodal values c_{e_i} the square-shaping of the pattern starts.

In further computations this process would continue, combined with a slowly increasing size of the zones with constant binodal values. We note that according to (5.4), at $c_m = \frac{1}{2}$ the D_2-term is entirely in charge of the development of large concentration fluctuations at the late process stages. Eventually, the concentration distribution is expected to approach a characteristic

stationary configuration, known as a cluster or two-zone structure. Such stationary solutions, however, are not stable since they are not local minima of the corresponding free energy functional (see section 3.4). On the other hand, stationary solutions with large wavelengths are quasi-stable (or kinetically stable) in the sense that they can exist for a long time, until a rapid change of pattern occurs leading to a new state with a slightly larger wavelength. This is a reason the system can be considered as being in permanent motion (with the ordinary time scale) towards the unique stable single-interface equilibrium.

As we could observe from the performed computations, in the case of equi-phase systems there is no similarity between the periodic pattern formed in time and the perturbation initially

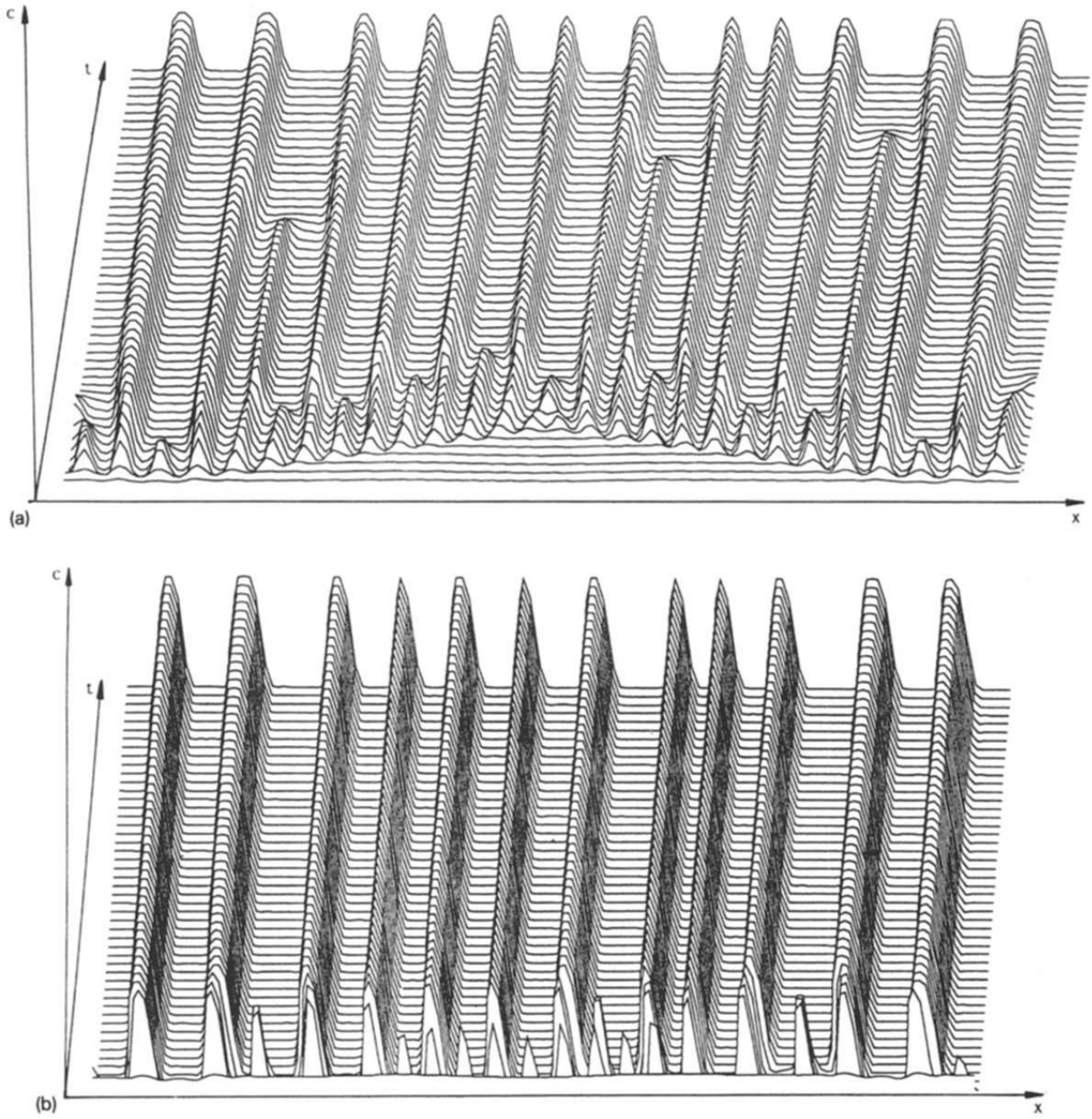


Fig. 4.

imposed around a spatially homogeneous state. However, as visualized in figs. 4–6, the influence of such perturbations becomes more apparent as the mean concentration c_m decreases. Fig. 4a displays an early stage of time development of the concentration perturbation in the case $c_m = 0.3$. In fig. 4b a continuation of this development for later times is shown. The number of modulations is smaller than in fig. 3. This remains in agreement with expression (3.33) for the fastest wavelength λ_{opt}. Actually, since $|D_0|$ increases in c_m, this contributes to an increase of λ_c and λ_{opt}. In contrast to the equi-phase case, the process development is now controlled by the D_1-term in (5.4). This explains the effect of zone aggregation at early stages of the process (see fig. 4a). Besides, coarsening rates are higher than those arising at $c_m = \frac{1}{2}$. A modulated phase pattern is approached (see fig. 4b).

Fig. 5 shows the time-development of phase separation in the system with $c_m = 0.2$, still within the spinodal region but already very close to the spinodal value $c_{s_1} = 0.17$. Now the impact of D_0 diminishes even more while other terms of D become dominating. In this case, the initial perturbation of concentration evidently affects its further evolution. The number of zones in fig. 5 is smaller than in the previous cases but periodicity is still preserved.

The time development of the concentration perturbation displayed in fig. 6 corresponds to the system with mean concentration $c_m = 0.15$ in the metastable region. Nucleation effects occur now, with peaks formed at points at which the initial perturbation is large enough. The amplitude of the peak in fig. 6 rapidly grows up to the binodal value. Then the peak increases in size and becomes rectangular.

The results of numerical experiments presented in figs. 3–6 are qualitatively agreeable with those obtained in [18,19] by using a Fourier type approach.

5.3. *Non-isothermal case*

The results of numerical simulation visualized in figs. 7, 8 refer to the non-isothermal developments. The system, defined on the interval $]0, L[$, is assumed to be cooled symmetrically at $x = 0$ and $x = L$, according to Newton's law

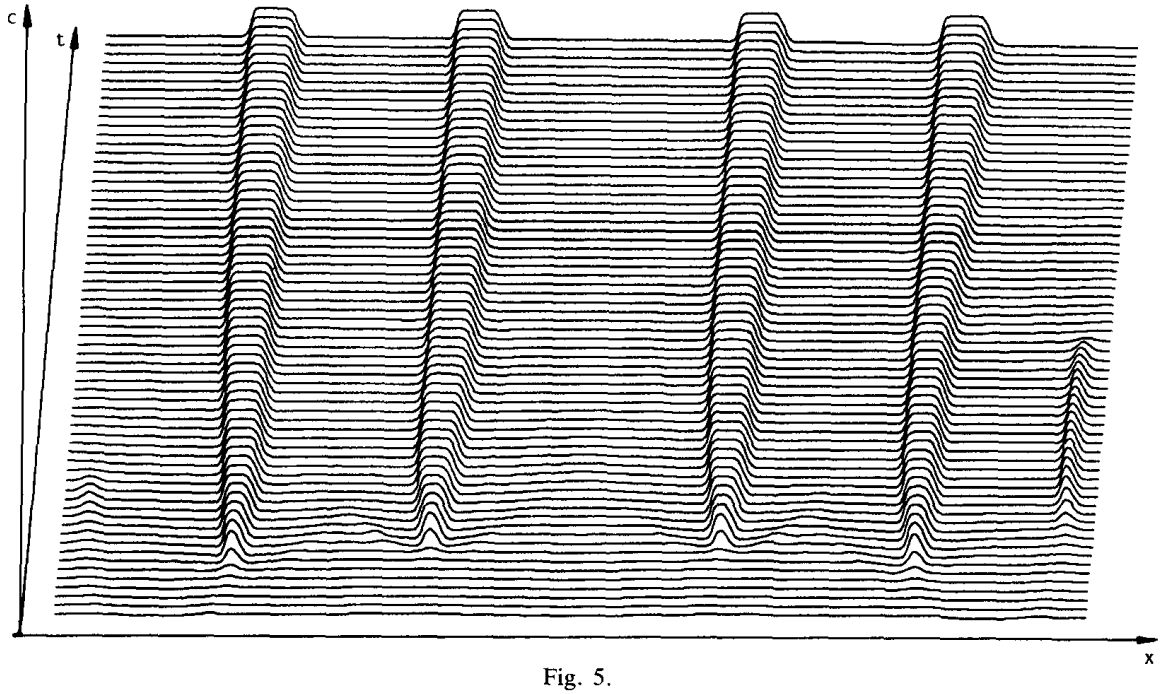


Fig. 5.

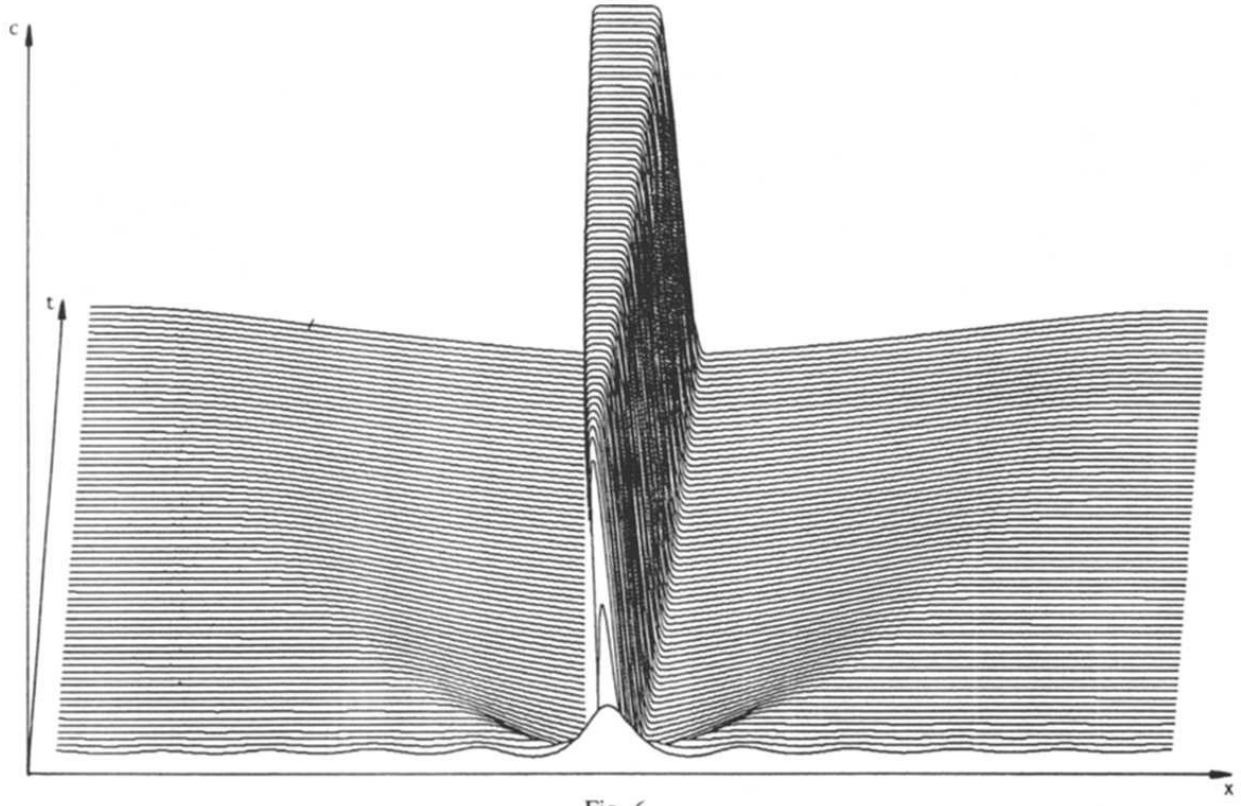


Fig. 6.

(2.13), (2.14). Thus, the computation can be restricted to the interval $[0, \frac{1}{2}L]$, with condition (2.13) at $x = 0$ and the homogeneous Neumann condition at $x = \frac{1}{2}L$.

It is essential for the dynamical development of the system that diffusion and heat transfer are phenomena of different time scales. Transient thermal developments are, as a rule, much faster compared to slow diffusion mode. Nevertheless, as the dynamics of thermal processes influences the development of spatial patterns, its significance is evident. To visualize this impact in numerical experiments, we have adjusted values of thermal parameters so as to diminish the gap between both time scales involved. The diffusion data are listed in section 5.2. The thermal parameters are

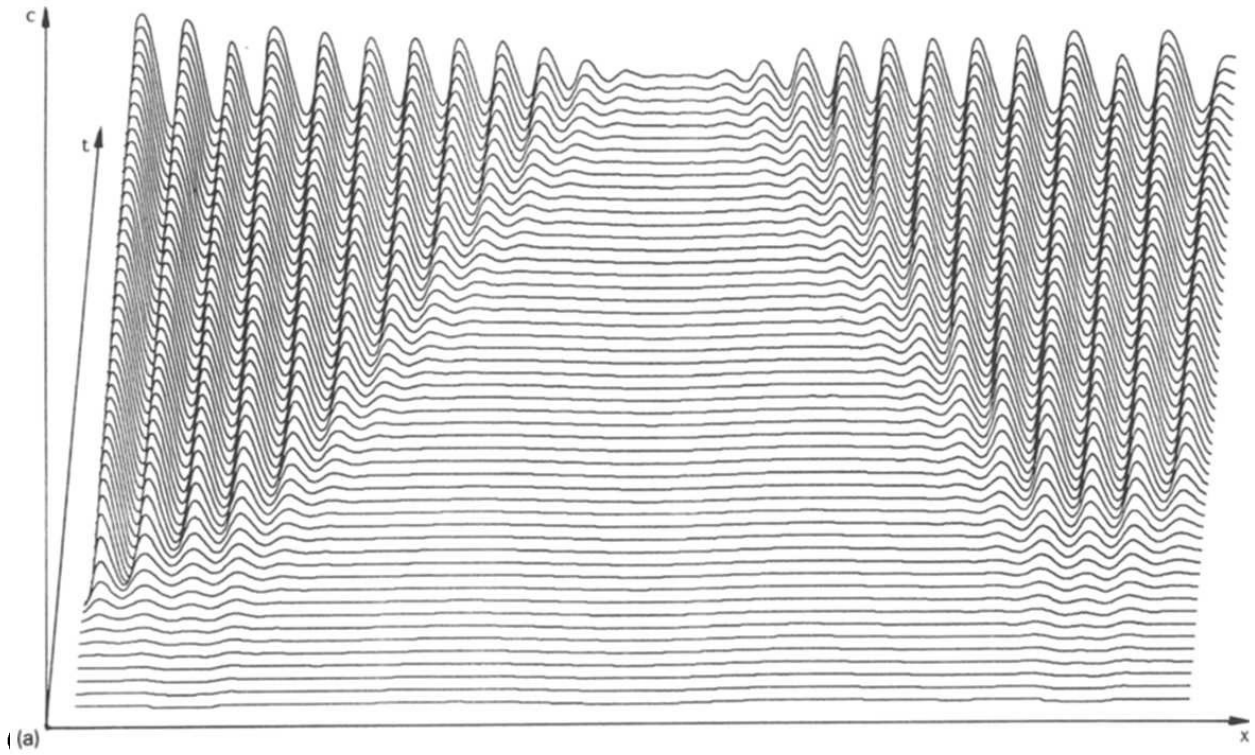
$$\delta_1 = 10^4\,, \quad \delta_2 = 10^4\,, \quad \delta_V = 10^{-4}\,,$$

$$l_{22} = 100\,, \quad l_{21} = 0\,,$$

$$g = 0\,, \quad g_b = 0\,, \quad p_0 = 1\,,$$

$$\theta_{crit} = 1500\,, \quad \theta_{ext} = 1000\,.$$

The initial perturbation of the homogeneous state c_m is given by (5.5) while the initial temperature is assumed constant, $\theta(0) = \theta_0 = 1200$, below the critical point $\theta_{crit} = 1500$. The external temperature is $\theta_{ext} = 1000$. The system with equiphase composition $c_m = \frac{1}{2}$ is considered. For the initial perturbation around c_m, its time development at early stages is shown in fig. 7a. A continuation of this development is displayed in fig. 7b. Because the initial temperature is below θ_c, the process starts to develop over the whole space interval. According to (3.31), all perturbations with a super-critical wavelengths are amplified. By (3.32) and (5.4), the fastest growing wave λ_{opt} is longer for smaller undercooling $|\theta - \theta_{crit}|$, see fig. 7a, corresponding to the initial temperature $\theta_0 = 1200$ versus the isothermal situation at $\theta = 1000$ depicted in fig. 3a. As visible in fig. 7a, at a very early stage the concentration pattern is dependent on its initial form.



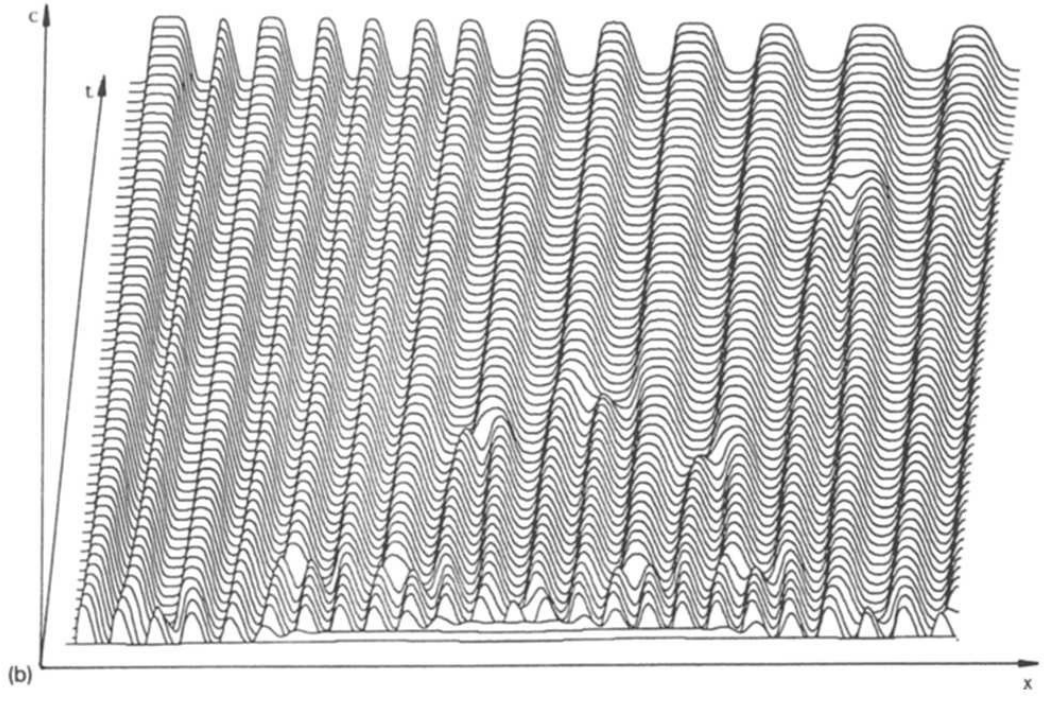


Fig. 7.

After some time, however, a modulated pattern with gradually increasing amplitude is propagating. The effect of zone aggregation can be observed in fig. 7b. At first this can be noticed close to the left boundary point where cooling is applied, then it occurs also in the center of the domain. As an additional effect, fig. 7b shows the reaction of the concentration pattern to the undercooling $\theta - \theta_{crit}$ (with $\theta < \theta_{crit}$). Since temperature near $x = 0$ is lower than near $x = \frac{1}{2}L$, the modulated structure has shorter period and larger amplitude in a neighbourhood of the left boundary point than in the center of the domain. This remains in a qualitative agreement with the expression (5.2) for the binodal values, and with (3.33) for the predominant growth wave.

Fig. 8a displays the temperature distribution at early stages of phase separation and fig. 8b its further development in time (time intervals in figs. 8a, 8b coincide with those in figs. 7a, 7b).

We see that the effects of phase grain formation and aggregation of zones lead to a considerable increase of temperature locally in space. Because of the boundary cooling temperature decreases globally over the whole domain. Asymptotically it will approach a stationary value.

The influence of the concentration pattern on temperature distribution can be approximately explained on the basis of the corresponding equations. Let us first consider an early stage of phase separation, associated with the growth of spatial heterogeneities. For f given by (5.1), (3.7), with $c_m = \frac{1}{2}$ and $\delta_2 = 0$, the internal energy E has the form

$$E(u, \theta) = \delta_V \theta - \alpha_1 \theta_{crit} u^2 + \alpha_6 u^6 + \tfrac{1}{2}\kappa |\nabla u|^2 . \tag{5.6}$$

According to the explicit form (3.31) of the solution, for sufficiently small β, that is, for perturbations u with large wavelengths,

$$u^2 \gg \tfrac{1}{2}\kappa |\nabla u|^2 .$$

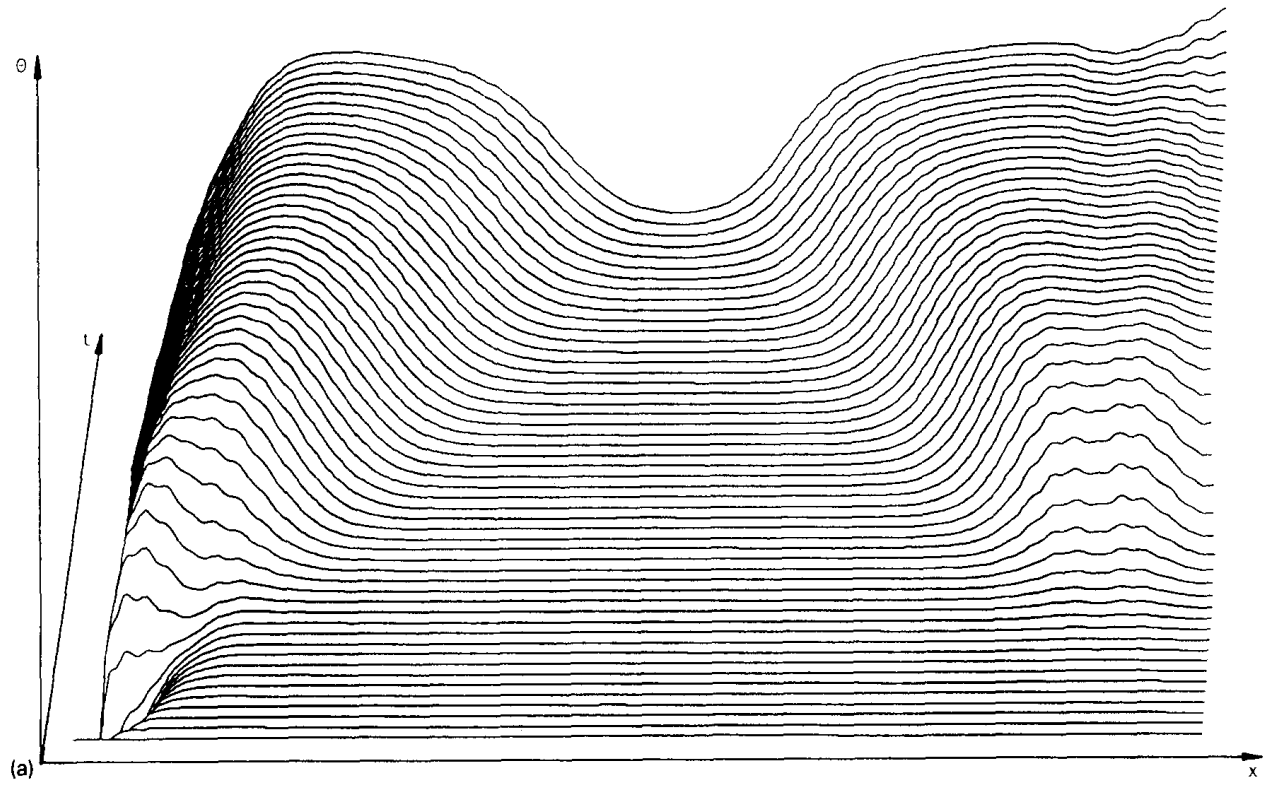


Fig. 8.

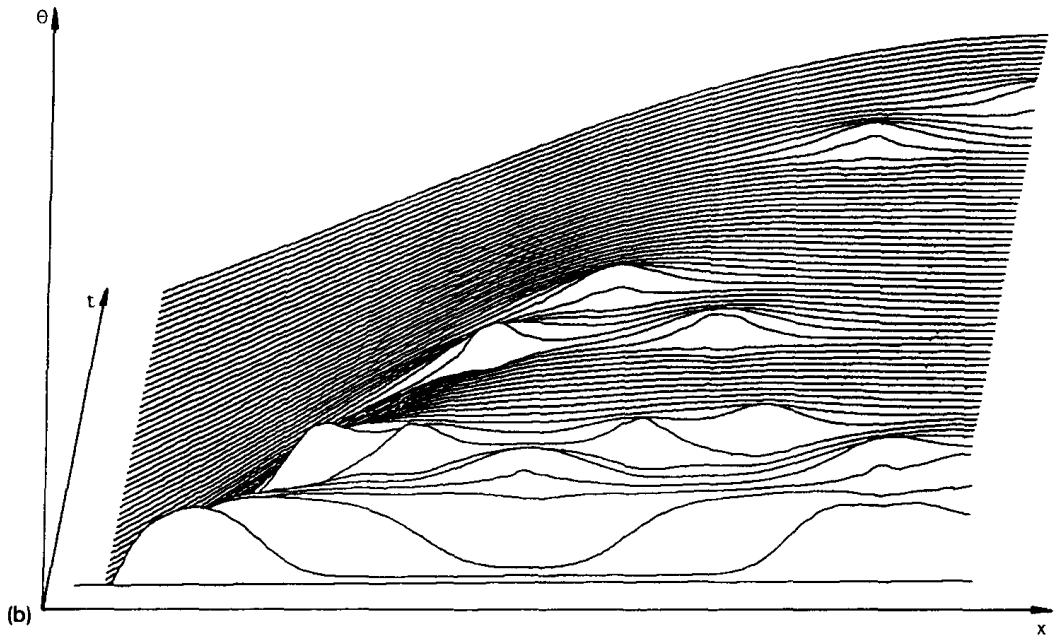


Fig. 8 (cont.).

Besides, for early stages of the process, the term $\alpha_6 u^6$ in (5.6) is small against $\alpha_1 \theta_{crit} u^2$, thus can be neglected. Then, over a short time interval, the internal energy can be approximated by

$$E(u, \theta) \approx \delta_V \theta - \alpha_1 \theta_{crit} u^2 .$$

Hence, using again the expression (3.31) for u, we conclude that

$$\partial_t E \approx \delta_V \, \partial_t \theta - 2\alpha_1 \theta_{crit} R(\beta) u^2 .$$

Assume β is small. Then $R(\beta)$ is positive and, due to cooling at the boundary, we can assume that $\partial_t \theta < 0$. With this, we infer that $\partial_t E < 0$. Hence, in view of the energy equation (2.24) (with $l_{22} = \text{const.}$, $l_{21} = 0$, and $g = 0$),

$$-\partial_t E - l_{22} \Delta w = 0 ,$$

it follows that $\Delta w > 0$. Therefore w must decrease, while θ is correspondingly increasing.

For late stages of phase separation the role of the κ-term of the internal energy (5.6) becomes dominant. Since the decay of phase grains leads to a decrease of the interfacial energy, $\partial_t E$ should be negative. Hence, as before we conclude that θ is increasing in space.

Altogether, the results of numerical experiments confirm physical relevance of the proposed model.

Remark

A different framework for thermodynamically consistent models of the kinetics of phase transitions was recently proposed by Penrose and Fife [25]. The derivation of their models is based on an entropy functional instead of the Landau–Ginzburg free energy functional.

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